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On the Effects of Boarding Congestion on Treatment Time in Emergency Departments

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We conduct multimethod research to examine the impact of boarding congestion on patient treatment time in emergency departments (EDs) and evaluate two operational strategies proposed to mitigate its adverse effects. First, we perform an empirical analysis using ED visit data from eight Canadian hospitals, identifying an inverted U-shaped relationship between boarding congestion and treatment time, where moderate congestion exacerbates delays. Mechanism testing reveals that ED resource use by boarding patients mediates this nonlinear effect, with hospitalist visits—by inpatient service physicians—moderating the relationship between congestion and ED resource use. Second, building on the empirical findings, we develop a simulation model to assess the two proposed strategies within the intricacies of real-world EDs. Simulation results show that these strategies can jointly reduce the adverse impact of boarding congestion on treatment-section congestion by approximately 44%, corresponding to 2.64 fewer patients in the treatment section on average and a reduction in average treatment time from 6.25 to 5.98 hours in a typical ED.

Key words: empirical study, healthcare operations management, emergency departments, patient flow control, ED hospitalist scheduling, ED boarding

1. Introduction

Emergency department (ED) boarding refers to the temporary holding of patients in the ED after a decision is made to admit them (American College of Emergency Physicians 2018). Various factors can cause long boarding times in EDs, including high ward occupancy (Powell et al. 2012), inefficiencies in the admission process (Mohr et al. 2020), and patient-flow-related issues such as delays in bed assignment, inpatient placement, and coordination of patient transfers from the ED to wards (Feizi et al. 2023). According to the Agency for Healthcare Research and Quality (2018) survey, 9 out of 10 US hospitals reported holding admitted patients in the ED, with boarding patients accounting for 10-20% of the total ED census (McKenna et al. 2019). ED boarding has been

associated with numerous negative consequences for boarding patients, including lower-quality care (Rabin et al. 2012), decreased patient safety (Armony et al. 2015), delays in receiving timely care (Sills et al. 2011), prolonged in-hospital stays (Chan et al. 2017), and reduced patient satisfaction (Viccellio et al. 2013).

In addition to its detrimental effect on patients' health conditions, ED boarding can have adverse effects on ED performance, affecting all stages of ED patient flow, from the waiting room and treatment section to the boarding area (Batt and Terwiesch 2017). Several studies have investigated the impact of boarding congestion on the length of stay (LOS)[1] in EDs (e.g., White et al. (2013), Khare et al. (2009b)). These studies demonstrate that boarding congestion increases the ED LOS for both patients admitted to the hospital from the ED and patients discharged directly from the ED. Some studies have also investigated the effect of boarding congestion on ED waiting time[2] (e.g., Hoot and Aronsky (2008), Saghafian et al. (2015)). These studies reveal that boarding congestion can prolong waiting times in the waiting room as boarding patients continue to occupy ED beds, thereby impeding the admission of new patients to the treatment section. However, to the best of our knowledge, prior studies have not investigated how overcrowding caused by boarding patients in emergency departments affects the time it takes to provide treatment to patients, nor have they explored the potential mechanisms underlying this relationship.

Understanding this impact and the underlying mechanisms can provide valuable insights into the effects of boarding congestion and facilitate the development of effective solutions to mitigate its potential negative impact. Therefore, this study aims to address the following questions:

- How does boarding congestion impact treatment time in EDs?
- What are the key mechanisms that contribute to the impact of boarding congestion on treatment times?
- How can we effectively mitigate the potential adverse impact of boarding congestion on treatment times?

ED boarding may influence the treatment time in multiple ways. First, boarding patients may increase the workload and multitasking level of ED staff. Although admitted patients are formally under the care of inpatient physicians, ED staff may still be required to provide medical attention to boarding patients (Armony et al. 2015). It is not surprising that, when boarding times are prolonged, these patients may require interim medical attention; thus, ED staff (e.g., physicians and nurses) may often need to provide inpatient-type care to these patients.[3] (Mohr et al. 2020).

[1] LOS refers to the duration of time a patient spends in the ED from the moment they are registered until they are either discharged or admitted to the hospital for further treatment (Yoon et al. 2003).

[2] Waiting time refers to the duration from when the patient is registered in the ED until the patient is transferred to the treatment section (Batt and Terwiesch 2017).

[3] Throughout the paper, "inpatient service" refers to the clinical specialty/team managing admitted patients, whereas "ward" refers to the physical hospital unit where patients are placed. The terms are used intentionally according to the operational context discussed.

Additionally, some boarding patients may require diagnostic tests while waiting to be admitted to the hospital (Coil et al. 2016). Prior literature shows that such an increase in care providers' workload and multitasking level can impact service throughput. For instance, Kc (2014) illustrates that while initial multitasking improves productivity, excessive multitasking decreases productivity due to increased non-value-added activities and task switching. Moreover, multitasking often results in increased interruptions (Chisholm et al. 2001), further deteriorating the service rate. All of these factors can contribute to longer treatment times for patients in the ED.

Second, although boarding congestion is only one contributor to overall ED crowding, it may still affect the discretionary behavior of care providers. Prior research shows that crowding in EDs increases physicians' cognitive load (Pines 2017), which can alter behaviors related to diagnostic test ordering and patient prioritization. In our setting, boarding congestion represents an important output-side source of pressure because boarding patients occupy ED beds and compete for provider attention and treatment resources. As a result, when boarding congestion increases, physicians may order more diagnostic tests to temporarily reduce workload while awaiting results (Berry Jaeker and Tucker 2019) and may also use additional testing to avoid unnecessary admissions, particularly for "gray area" patients whose appropriate disposition remains ambiguous (Soltani et al. 2022). In addition, under higher workload, care providers may adjust their prioritization schemes to balance the needs of boarding patients and emergency patients[4] (Armony et al. 2015). In particular, higher congestion may lead care providers to favor easier tasks, consistent with "task completion preference" (Kc et al. 2020). Therefore, when evaluating the impact of boarding congestion on treatment times, it is important to consider its effect on the added workload for ED resources.

We answer our research questions in two steps. First, using a large dataset encompassing 470,173 ED visits across eight hospitals, we conduct an empirical analysis to investigate the impact of boarding congestion on ED patient treatment time. Our analysis reveals an inverted U-shaped relationship between boarding congestion and treatment time. For instance, we find that the mean treatment time increases by 19%, from 216 minutes to 257 minutes, when boarding congestion increases from 0.1 to 0.4 (i.e., boarding capacity utilization increases from 10% to 40%). However, as congestion further increases from 0.4 to 0.7 (i.e., utilization rising from 40% to 70%), the mean treatment time decreases from 257 to 245 minutes, representing a 4.7% decrease.

We extend our empirical analysis to examine the mechanism underlying the non-linear effect of boarding congestion on ED treatment time. Inpatient capacity constraints and administrative delays in bed assignment and transfers often keep boarding patients in the ED for extended periods, during which they continue to require care. We posit that ED resource utilization by boarding

[4] We refer to patients as "emergency patients" while they are in the waiting room or treatment section and "boarding patients" while they are in the boarding section.

patients mediates the effect of boarding congestion on treatment time, while hospitalist visits–by inpatient service physicians accepting admitted patients from the ED–moderate the relationship between boarding congestion and ED resource use. At low congestion levels, boarding patients are primarily managed by ED staff, increasing the use of physician and nursing time as well as diagnostic services (laboratory tests and imaging). This intensifies ED capacity constraints and prolongs treatment times, generating the upward-sloping portion of the inverted U-shaped relationship. At high congestion levels, hospitalist involvement is more likely, shifting care responsibilities away from ED staff and reducing reliance on diagnostic services, as hospitalists order tests more selectively. Consequently, boarding patients' dependence on ED resources declines, capacity constraints ease, and ED treatment times shorten, producing the downward-sloping portion of the inverted U-shaped relationship. Our empirical results show that this mechanism serves as a key mediator of the inverted U-shaped relationship between boarding congestion and ED treatment time.

In the second step, motivated by our empirical findings, we propose two interventions to mitigate the impact of boarding congestion on treatment time: (i) prioritizing patients in the treatment section and (ii) implementing operational policies for hospitalist visits to the boarding section.

Regarding the first intervention, ED physicians at our partner hospitals currently prioritize boarding-patient checkups regardless of how congested the treatment section is. Therefore, our goal is to examine whether this rule should instead be adjusted dynamically based on congestion in both areas. We propose an easy-to-implement threshold-based patient prioritization policy that allocates ED physician attention between a treatment queue and a boarding checkup queue based on their congestion. Under this policy, treatment patients are prioritized when congestion in the treatment section exceeds a threshold, while boarding patients regain priority when the checkup queue becomes sufficiently large, protecting treatment throughput while preventing excessive delays for boarding patients. To evaluate the policy, we calibrate a simulation model using our data: the results show that the policy reduces the adverse impact of boarding congestion on treatment section congestion by 37.48%, corresponding to an average reduction of 2.23 patients in the treatment section.

Our second intervention focuses on hospitalist visits to boarding patients in the ED. In our setting, hospitalists are inpatient service physicians responsible for accepting patients admitted from the ED, coordinating their transfer to the appropriate inpatient service, and providing subsequent inpatient care. Because admitted patients may remain in the ED while awaiting inpatient beds, hospitalists may visit these boarding patients in the ED to assume or coordinate their care before transfer. (Kathuria et al. 2010) We refer to these encounters as hospitalist visits. Although hospitalist visits are an established practice for addressing ED boarding (Howell et al. 2010), prior research has not examined the operational challenges associated with scheduling these visits, particularly how to provide effective care to boarding patients without substantially increasing hospitalists' workload.

Discussions with staff members at our partner hospitals revealed that visit practices vary across hospitals and inpatient services, even within the same hospital, due to the absence of standardized protocols. To address this issue, we propose two hospitalist visit scheduling (HVS) policies, namely time-based and census-based policies, that aim to reduce the workload imposed on ED resources by boarding patients while limiting additional burden on hospitalists (Zhu 2018).

Our simulation results show that both HVS policies mitigate the adverse impact of boarding patients on treatment–section congestion, with the census-based policy achieving greater improvement because it responds more directly to boarding demand. Specifically, the census-based policy reduces this adverse impact by 31.4%, corresponding to an average reduction of 1.87 patients in the treatment section. The performance of both HVS policies, however, depends on ward congestion. When inpatient wards are highly occupied, hospitalists have less flexibility to leave their wards, making scheduled visits to boarding patients less likely to occur. Consequently, the benefits of HVS diminish as inpatient congestion becomes more severe. This decline is especially pronounced for the census-based HVS policy because it tends to schedule visits during periods of high boarding demand, which often coincide with periods of high ward congestion and limited hospitalist flexibility to visit the ED.

Finally, we examine the joint implementation of the patient-prioritization policy and the census-based HVS policy. Among all interventions considered, the joint implementation provides the best overall performance. Specifically, we find that the combined intervention reduces the adverse impact of boarding patients on treatment-section congestion by 44.37%, corresponding to an average reduction of 2.64 patients in the treatment section, while also improving treatment time from 6.25 hours to 5.98 hours.

2. Literature Review

Our work relates to two streams of research: empirical studies of the impact of crowding on service times, and studies of boarding patients, patient flow control, and the role of hospitalists in ED performance.

2.1. The Impact of Crowding on Service Time

Fundamental queuing theory traditionally assumes that service times remain fixed regardless of system congestion (Wolff 1989). However, recent empirical studies in operations management have demonstrated that crowding can have an impact on service time across various sectors, including banking (Staats and Gino 2012), hospitality (Tan and Netessine 2014), and retail (Wang and Zhou 2018). In the healthcare literature, the impact of crowding on service time has received considerable attention, likely due to the criticality of service time in healthcare systems and the limited control

of overcrowding within these systems. Prior literature has investigated the effects of crowding measures on service time in different healthcare settings, including intensive care units (ICUs) (Kim et al. 2017), hospital wards (Berry Jaeker and Tucker 2017), and patient transport services and cardiothoracic surgeries (Kc and Terwiesch 2009).

In the context of EDs, several studies examine the effects of crowding at different stages of the ED on treatment time (e.g., Batt and Terwiesch (2017), Kc (2014)). For instance, Batt and Terwiesch (2017) investigate the impact of waiting room census on treatment time and identify an inverted U-shaped relationship. They attribute this relationship to early task initiation, delay in medication delivery, and nurse rushing. Moreover, they show that the treatment time increases with a higher treatment census, which can be explained by ED staff multitasking caused by overcrowding in the treatment section (Kc 2014). Additionally, various medical studies evaluate the impact of ED crowding (including crowding in the waiting room, treatment section, and boarding section) on LOS and waiting time (e.g., McCarthy et al. (2009), Timm et al. (2008), Lucas et al. (2009)). These studies, employing statistical analysis, consistently report that ED crowding leads to increased LOS and waiting time (e.g., Hoot and Aronsky (2008), Morley et al. (2018)).

There are a limited number of medical studies investigating the effects of ED boarding on LOS and waiting times (e.g., White et al. (2013), Khare et al. (2009b), Carmen et al. (2018)). These studies consistently demonstrate that boarding congestion contributes to prolonged LOS and waiting times, as patients awaiting admission occupy valuable ED beds and hinder the smooth movement of new patients to the treatment section. Recent medical studies have shed light on how boarding affects EDs and the overall healthcare system. For example, Lee et al. (2022) show that implementing a boarding restriction protocol significantly reduces crowding in EDs. This not only helps reduce occupancy levels but also shortens the LOS for admitted patients by about 20%. Similarly, Laam et al. (2021) find that for every additional patient boarded, the median ED LOS for all admitted patients increases by roughly 12 to 14 minutes. This shows how boarding can create ripple effects that impact the entire system. Moreover, some studies have linked prolonged boarding to longer inpatient stays and a higher risk of mortality, though the evidence on mortality varies across different settings (Forster et al. 2003, Boulain et al. 2020, Boudi et al. 2020, Howlett et al. 2026, Yousufuddin et al. 2026). Despite extensive evidence linking boarding congestion to increased LOS and waiting times across both operations management and medical literature, its impact on treatment time and the underlying mechanisms remain largely unexplored. In the operations management literature, there are only two exceptions: First, Carmen et al. (2018) analytically show that boarding patients adversely affect ED congestion and maximum throughput. Second, based on data from a large Israeli hospital, Armony et al. (2015) observe that a boarding patient requires an average of 1.5 minutes

of the ED physician's time every 15 minutes. However, this study did not empirically examine the impact of boarding patients on ED performance.

Similarly, in the medical literature, there are only a few studies demonstrating that hospital occupancy prolongs LOS in EDs and negatively impacts ED performance (e.g., Hillier et al. (2009)). However, it is essential to examine the impact of boarding congestion separately from hospital occupancy. Although hospital occupancy is recognized as a critical factor contributing to boarding congestion (Powell et al. 2012), other factors such as inefficient admission processes and patient-flow issues also contribute to boarding congestion (Luo et al. 2013, Feizi et al. 2023). Moreover, it is important to acknowledge that the effects of boarding congestion and hospital occupancy on the ED staff may differ since hospital occupancy is not directly observable to the ED staff.

Therefore, our research provides the first empirical evidence of the impact of congestion caused by boarding patients on ED treatment time. We provide empirical evidence demonstrating that the utilization of ED resources by boarding patients is one of the mechanisms underlying the relationship between boarding congestion and treatment time. Building on these findings, we propose two interventions to manage boarding patients and mitigate the negative impact of boarding congestion on ED performance: (i) patient flow control in the treatment section and (ii) scheduling of hospitalist visits. In the following sections, we provide a review of the existing literature on these two interventions. This gap motivates a deeper examination of how boarding congestion affects service processes within the ED, particularly at the treatment stage.

2.2. Patient Flow Control and Hospitalists

The first intervention is to control boarding patient flow in EDs by allocating ED resources judiciously between boarding patients requiring the attention of the ED staff and other ED patients. This intervention addresses an important gap in the literature, as existing patient flow control models largely abstract away from the operational role of boarding patients, despite their significant impact on ED congestion and resource utilization. Most studies on patient prioritization in EDs aim to balance the work of ED physicians based on two classes of patients: new patients after triage and in-progress patients (e.g., He et al. (2019)). Other classifications of patients in EDs for the patient flow control problem are based on predicting patients' dispositions (e.g., Saghafian et al. (2012)) and the complexity of treatment procedures (e.g., Saghafian et al. (2014)) (more details on customer classifications in hospital and other healthcare systems can be found in (Helm et al. 2011, Ranjan et al. 2017)). Most existing research on prioritizing patients in EDs assumes that patients do not need the ED staff's attention during boarding time. One exception is the study conducted by De Boeck et al. (2019), which compares four ad-hoc priority policies to balance the ED physician's workload on boarding and other ED patients using simulation. They assume that boarding patients

waiting to be checked up again by ED physicians cannot move to the ward before being evaluated by the ED physician. We relax this assumption in our simulation model, which is consistent with our observations in the study EDs. As a result, existing prioritization frameworks may overlook congestion effects by failing to explicitly consider the resource demands and operational externalities linked to boarding patients.

The second intervention proposes a planning framework for hospitalist visits in the boarding section, which directly addresses the coordination challenges arising from prolonged boarding. Some medical studies report the benefits of using hospitalists to manage admitted patients who remain boarding in the ED to mitigate the negative impacts of prolonged boarding time and improve ED performance in practice (e.g., Hrycko et al. (2019)). However, there is a lack of prior OM/OR literature focusing on this role in EDs, particularly regarding scheduling policies. Our study is the first to provide a scheduling framework for ED hospitalist visits.

Overall, the literature on patient flow control and the role of hospitalists shows how crucial operational coordination is for reducing congestion in EDs. However, many existing studies do not specifically consider the dynamics of boarding, nor do they look at how various interventions can affect treatment times by influencing resource use and provider behavior. This gap motivates our focus on jointly modeling patient prioritization and hospitalist scheduling while considering the impact of boarding congestion.

3. Empirical Analysis

In this section, we start by providing an overview of the clinical setting and describing our data. Following that, we introduce our identification strategy to quantify the impact of boarding congestion on treatment time and present our estimation results.

3.1. Clinical Context

The patient flow in the study EDs is consistent with that described in prior literature (e.g., Batt and Terwiesch 2017, Soltani et al. 2022). Figure 1 illustrates this flow. Upon arrival, patients are first assessed by a triage nurse, who records vital signs and assigns a triage code according to the Canadian Triage and Acuity Scale (CTAS), reflecting the urgency of care. Following triage, patients wait in a waiting area until they are called to a treatment room. There, an ED physician examines the patient, determines the appropriate diagnostic and treatment procedures, initiates treatment or orders medications, laboratory tests, or diagnostic imaging. The ED physician may reassess patients as laboratory results or diagnostic images become available and may subsequently order additional tests or modify treatment. Thus, the treatment process consists of a sequence of interdependent clinical and operational activities rather than a single continuous service encounter. At the end of the treatment stage, a disposition decision is made regarding discharge or hospital admission.

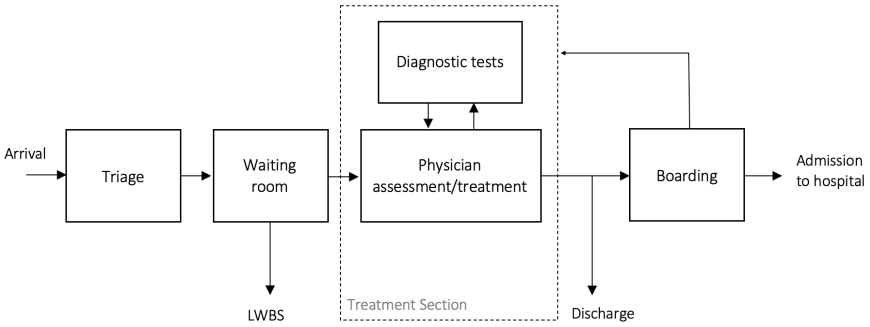


Figure 1 Typical Patient Flow

For patients requiring admission, treatment in the ED is followed by a boarding stage, during which patients remain in the ED while awaiting assignment to a bed in the appropriate ward. Each inpatient service is staffed by hospitalists—physicians who specialize in inpatient care–who are responsible for accepting admitted patients from the ED, coordinating their transfer to the ward, and providing subsequent care. Due to inpatient unit capacity constraints and administrative delays in bed assignment and transfers, boarding patients may remain in the boarding area for extended periods. Although boarding patients are formally under the care of hospitalists, hospitalists are not continuously present in the ED and do not visit boarded patients frequently. Moreover, the study hospitals lack standardized protocols governing the timing or frequency of hospitalist visits. Instead, hospitalists are notified of the number of patients boarding on their service and independently decide when to visit these patients while balancing other inpatient responsibilities. Therefore, during those extended periods, boarding patients may continue to rely on ED resources: ED physicians may provide ongoing assessment and monitoring and order additional diagnostic tests; ED nurses monitor boarding patients and support their ongoing care; laboratory and imaging services process these diagnostic orders; and ED treatment beds provide the physical capacity in which boarding patients undergo assessment, treatment, and monitoring. Consequently, boarding patients continue to consume both physical capacity and clinical resources that are also required for patients undergoing active treatment, creating competition for treatment beds, physician and nursing attention, and diagnostic resources and potentially leading to operational delays for other ED patients.

3.2. Data and Unit of Analysis

Our data come from eight hospitals located in a metropolitan area in Canada, comprising three secondary hospitals and five tertiary hospitals. Compared to secondary hospitals, tertiary hospitals are generally larger facilities with more specialized equipment and inpatient services (Flegel 2015). Despite these differences, patient flow in the EDs is similar across both hospital types.

We obtained patient-level ED visit records from all eight hospitals over an 18-month period. Each record contains detailed timestamps for patient arrival, triage, ED bed assignment, and disposition, as well as information on diagnostic tests, mode of arrival (e.g., walk-ins, ambulance), patient demographics (e.g., age, gender), chief complaint, and triage level as measured by the Canadian Triage and Acuity Scale (CTAS). As described in Appendix A1, we apply a set of sample restrictions to focus on visits that follow the standard ED care process, resulting in a final analysis sample of 470,173 ED patient visits. The unit of analysis is an ED visit-hospital pair, indexed by i and h, respectively.

3.2.1. Dependent Variable: In our analysis, the primary dependent variable is treatment time. Consistent with prior research (e.g., Batt and Terwiesch 2017), we define treatment time for ED visit i at hospital h ($TreatmentTime_{ih}$) as the elapsed time, in minutes, from when the patient is called to the treatment section until the physician makes the disposition decision.

3.2.2. Key Independent Variable: The key independent variable in our study is boarding congestion during the treatment time of ED visit i at hospital h, denoted as $Congestion_{ih}$. To operationalize boarding congestion, we follow Freeman et al. (2021) and define it as the ratio of boarding census to boarding capacity. Drawing on previous studies (e.g., Batt and Terwiesch 2017, Chan et al. 2017), we measure boarding census as the average hourly number of boarding patients–those who have received an admission disposition but have not yet been transferred to the ward–during the treatment time of ED visit i. Boarding capacity is defined, following Freeman et al. (2021), as the 95th percentile of the hourly boarding census for each ED. Unlike Freeman et al. (2021), who allow boarding capacity to vary over time to account for dynamic staffing and operational adjustments, the EDs in our study do not adjust boarding capacity. Accordingly, we treat boarding capacity as constant within each ED. Importantly, we use the full dataset–including observations excluded from the final analysis sample–to calculate both boarding census and capacity.

Two potential concerns with measuring boarding congestion as an average over the treatment period are its mechanical correlation with $TreatmentTime$, which determines the averaging window and the use of congestion information realized after treatment onset. These concerns are less pronounced in our setting because congestion is highly stable within treatment episodes. Specifically, congestion remains highly correlated (i.e., above 0.9) throughout the treatment period, and its dispersion remains stable as the treatment window lengthens. Consistent with this evidence, a variance-components decomposition shows that 91.5% of the variation in congestion occurs between visits, whereas only 8.5% occurs within visits. Moreover, as detailed in Appendix A2.1, our results are robust to measuring congestion at the treatment onset, which avoids both concerns, providing evidence that neither the mechanical correlation nor the use of future congestion realizations is

likely to drive our results. Consequently, we use the average boarding congestion in our analysis as it provides a parsimonious measure of the overall congestion environment experienced during the treatment period, consistent with the theoretical mechanism in Section 4.1.

3.2.3. Control Variables: We include several control variables that are potentially related to the dependent variable. Following Batt and Terwiesch (2017), we control for patient-level factors using five variables. $Gender_{ih}$ and Age_{ih} indicate the gender and the age of the patient in ED visit i at hospital h, respectively. $ArrMode_{ih}$ indicates whether the patient arrived as a walk-in or was brought by ambulance. $Triage_{ih}$ captures the patient's triage level based on CTAS, which ranges from one (most severe) to five (least severe). Finally, $DiagCode_{ih}$ is an indicator variable for the diagnostic disease code assigned to the patient.

We control for ED-level operational conditions using five variables. Prior research suggests that treatment time may be influenced by congestion in the waiting room, congestion in the treatment section, ED physician workload, and the staffing team providing care (Batt and Terwiesch 2017, Kim et al. 2015). To account for potential congestion in the waiting room and treatment section, we include $WaitCensus_{ih}$ and $TreatmentCensus_{ih}$, which represent the number of patients in the waiting room and in the treatment section, respectively, normalized by the number of licensed beds in each ED. To account for physician workload, we include $Workload_{ih}$, defined as the ratio of $TreatmentCensus_{ih}$ to the number of available ED physicians. These three variables are measured at the onset of treatment for the patient in ED visit i at hospital h. Consistent with Batt and Terwiesch (2017) and Soltani et al. (2022), we also include the quadratic terms for these variables to capture potential nonlinear effects. The EDs in our study follow rigid staffing plans, such that the staffing levels for a given shift and day of the week are highly consistent, with minimal variation due to individual staffing needs. We therefore control for staffing effects using shift fixed effects ($Shift_{ih}$), which identify the staffing team responsible for the patient in ED visit i at hospital h. Lastly, $AvgTriage_{ih}$ represents the average triage level of patients concurrently undergoing treatment and captures the overall complexity of care faced by ED staff during the treatment time of the patient in ED visit i at hospital h.

At the hospital level, we include two additional control variables. EDs share critical resources (e.g., diagnostic imaging equipment) with inpatient units, creating competition between ED patients and inpatients for these resources. Increases in inpatient demand may therefore affect ED treatment times (McCarthy et al. 2009, Allon et al. 2013). We control for inpatient demand using hospital occupancy ($HospOccupancy_{ih}$), measured at the start of the patient's treatment in ED visit i as the ratio of the number of patients in hospital h to its standard capacity. We also include hospital fixed effects ($Hospital_h$) to account for unobserved heterogeneity across the eight hospitals.

Lastly, we control for seasonality and time-of-day effects using indicator variables for year ($Year_{ih}$), month ($Month_{ih}$), day (Day_{ih}), and hour ($Hour_{ih}$), based on the start time of treatment for the patient in ED visit i at hospital h. We also include an indicator for weekend visits ($Weekend_{ih}$) and its interaction with $Hour_{ih}$ to capture differential temporal patterns on weekends. Descriptive statistics for all continuous and dummy variables are reported in Appendix A1. With a variance inflation factor score mean of 1.54 and a range between 1.00 and 3.52, there is little evidence of multicollinearity.

3.3. Identification Strategy

To examine the effect of boarding congestion on treatment time, we employ an accelerated-failure-time (AFT) model, a class of parametric duration models (Greene 2012). Duration models are designed to analyze the length of time until an event occurs–here, the completion of ED treatment–conditional on a set of covariates (Greene 2012). The AFT specification models the logarithm of treatment time as a linear function of explanatory variables (Greene 2012). This approach has previously been used in prior studies of ED treatment times (e.g., Batt and Terwiesch 2017). We specify the model as follows:

$$ln(TreatmentTime_{ih}) = \beta_0 + \beta_1 Congestion_{ih} + \beta_2 Congestion_{ih}^2 + \mathbf{Controls}_{ih}\boldsymbol{\beta} + \epsilon_{ih} \qquad (1)$$

where ϵ_{ih} is a random error term and $\mathbf{Controls}_{ih}$ is a vector of patient-, ED-, and hospital-level control variables, along with seasonality and time-of-day fixed effects. To accommodate a potentially nonmonotonic relationship between boarding congestion and treatment time, we include both the linear and quadratic terms of $Congestion_{ih}$. The coefficients β_1 and β_2 capture the marginal effects of boarding congestion on treatment time.

In the AFT framework, the assumed distribution of the error term determines the functional form of the survival model. We find that the Generalized Gamma distribution best fits the data (Bayesian information criterion (BIC)=1,544,399) and thus specify ϵ_{ih} to follow a Gamma distribution.

Endogeneity and Instrumental Variable: The estimates from Equation 1 may be subject to omitted variable bias arising from unobserved factors that jointly affect both *TreatmentTime* and *Congestion*. Boarding congestion in the ED is jointly determined by two processes: the inflow of patients admitted from the ED into the boarding area and the outflow of boarded patients to their designated inpatient services by hospitalists, reflecting the balance between admission and transfer rates. As described in Section 3.2, our analysis includes a rich set of control variables capturing patient volume, staffing configurations, physician workload, and the complexity of care faced by ED staff, all of which are closely related to treatment time. Because treatment time is associated with the rate at which patients are admitted from the ED into the boarding area–as shown in

the correlation matrix in Appendix A1–these controls are also correlated with key determinants of the inflow component of boarding congestion, reducing concerns about unobserved confounding on that margin. However, we are unable to observe several factors that influence the outflow from the boarding area to wards and that may simultaneously affect ED treatment time, making omitted variable bias more likely to arise from unobserved determinants of the outflow process.

To address this concern, we employ the control function (CF) approach. The CF method is a full-information likelihood framework that has been widely used in AFT models, and more broadly, in nonlinear models with endogenous regressors (e.g., Terza 2018, Soltani et al. 2022, Kim et al. 2015). Similar to two-stage least squares in linear models, the CF approach proceeds in two stages. In the first stage, we estimate the following linear regression model:

$$Congestion_{ih} = \theta_0 + \theta_1 Z_{ih} + \mathbf{Controls}_{ih}\boldsymbol{\theta} + v_{ih} \tag{2}$$

where Z_{ih} denotes an instrumental variable and v_{ih} is an error term capturing the endogenous component of $Congestion$. In the second stage, we augment Equation 1 to account for endogeneity:

$$ln(TreatmentTime_{ih}) = \alpha + \beta_1 Congestion_{ih} + \beta_2 Congestion_{ih}^2 + \mathbf{Controls}_{ih}\boldsymbol{\beta} + \gamma\hat{v}_{ih} + \epsilon_{ih} \tag{3}$$

where $\hat{v}_{ih}$ is the residuals obtained from the first-stage regression. Consistent with Wooldridge (2015), it is sufficient to estimate the first stage only for $Congestion_{ih}$, as the resulting residuals also correct for the endogeneity of its quadratic transformation. In all estimations, we use robust standard errors clustered at the hospital level.

Identification under the CF approach requires at least one instrument Z_{ih} that predicts boarding congestion while affecting ED treatment time only through its impact on $Congestion_{ih}$. Formally, the instrument must satisfy two conditions: (i) it must be strongly correlated with $Congestion_{ih}$ (i.e., the relevance assumption), and (ii) it must have no direct effect on $TreatmentTime_{ih}$ other than through $Congestion_{ih}$ (i.e., the exogeneity assumption).

As our instrument, we exploit regional demand shocks for inpatient services by incorporating additional data on daily utilization of inpatient services from eight nearby hospitals (four secondary and four tertiary hospitals) located in close geographic proximity to our eight focal hospitals. For each inpatient service in our sample, we identify the nearest corresponding inpatient service among these nearby hospitals. Utilization at the nearest inpatient service captures regional demand conditions for that service and is therefore predictive of utilization at the focal inpatient service. As confirmed by our partner hospitals and documented in Song et al. (2020), higher inpatient service utilization can impede the transfer of patients from the boarding area—through mechanisms such as reduced hospitalist availability or delayed bed assignment—thereby increasing boarding congestion (the relevance assumption). Importantly, unlike inpatient services within the same hospital,

inpatient services at nearby hospitals do not share resources with the ED of the focal hospital, as confirmed by the partner hospitals. Moreover, our instrument is based on utilization of a specific inpatient service at a nearby hospital rather than on the nearby hospital's ED census, ED capacity, or ambulance-diversion status. Thus, the instrument captures downstream inpatient capacity conditions rather than contemporaneous demand conditions in nearby EDs. In addition, utilization of these inpatient services is not information used by ambulance personnel or dispatchers to determine patient routing or diversion decisions at the focal hospitals. Consequently, utilization at the nearest inpatient service is unlikely to directly affect ED treatment times at the focal hospital through resource-sharing, patient-routing, or ED-demand channels. We further examine in Appendix A2.5 the plausibility of common regional demand shocks that can simultaneously influence the instrument and the treatment time through channels other than boarding congestion and find no evidence that such shocks affect our estimates. Taken together, these institutional and empirical considerations provide support for the exogeneity assumption. This instrument is conceptually analogous to a Hausman-type instrument that exploits cross-market (here, cross-hospital) variation, a strategy commonly employed in the operations management literature (e.g., Fisher et al. 2018, Kaushik and Gokpinar 2023).

Building on this theoretical foundation, we first identify the designated inpatient service for each patient in our dataset. For patients admitted to the hospital, the designated inpatient service is directly observed. For patients discharged from the ED, we impute their counterfactual designated inpatient service using diagnostic disease codes. Because patients sharing the same diagnostic code may be admitted to different inpatient services depending on treatment decisions, for each diagnostic code, we identify the most frequently observed designated inpatient service among admitted patients with that code, and use this as the counterfactual designation for discharged patients. With this, the average distance to the nearest inpatient service corresponding to a designated inpatient service in our sample is 4.2 miles (minimum 0.7 miles; maximum 13.3 miles), with 90% of observations having a nearest inpatient service within 6.7 miles. We then construct the instrument, denoted as $NearestIP_{ih}$, defined as the ratio of occupied beds to total beds in the nearest inpatient service corresponding to the patient's designated inpatient service on the day treatment begins for ED visit i at hospital h.

3.4. Results

We next present our empirical findings. We start by reporting the first-stage estimation results (Equation 2), along with formal statistical tests assessing the relevance and strength of the instrumental variable, to alleviate concerns about weak identification. We then report estimates from the AFT specification both using the model that does not account for potential endogeneity (Equation 1) and using the control function approach with our proposed instrument (Equation 3).

3.4.1. Instrument and First-Stage Estimation: Consistent with our identifying assumptions, $NearestIP_{ih}$ exhibits a correlation of 0.345 with $Congestion_{ih}$ and a near-zero correlation of -0.009 with $TreatmentTime_{ih}$, providing initial empirical support that the instrument is predictive of boarding congestion while plausibly orthogonal to ED treatment time.

As a falsification test, we construct an analogous "placebo" instrument using a separate set of eight hospitals located, on average, 110.9 miles (minimum 39 miles; maximum 169 miles) away from the focal hospitals. Because these distant hospitals are unlikely to be exposed to the same regional inpatient demand shocks, the identifying variation underlying the instrument should no longer be relevant. Consistent with this intuition, the placebo instrument exhibits a near-zero correlation with boarding congestion (–0.033). This attenuation in predictive power provides further support that the original instrument captures localized demand-driven congestion shocks rather than spurious correlations.

The first-stage regression results are presented in Table 1, column (1). The coefficient on $NearestIP$ is positive and statistically significant ($\theta_1 = 0.20$, $p < 0.001$), indicating that higher utilization at the nearest inpatient service is positively associated with ED boarding congestion at the focal hospital. The first-stage F-statistic ($F = 1108.28$, $p < 0.001$) far exceeds the conventional threshold of 10, confirming the relevance of the instrument. As expected, we also find that ED boarding congestion is positively associated with hospital occupancy, and is more pronounced under high physician workload and low congestion in the treatment section.

3.4.2. AFT Model: Table 1 presents the estimates from the baseline AFT specification (column 2), which does not yet correct for the potential endogeneity of boarding congestion. The results reveal a nonlinear relationship between $Congestion$ and $TreatmentTime$, with a positive linear term ($\beta_1 = 0.65$, $p < 0.001$) and a negative quadratic term ($\beta_2 = -1.04$, $p < 0.001$), consistent with a concave pattern.

Column (3) of Table 1 reports the results from the control function specification. The estimated coefficient on the first-stage residual is negative and statistically significant ($\gamma = -0.85$, $p < 0.001$), indicating the presence of unobserved factors that jointly affect boarding congestion and ED treatment time. A Wald test on the residual term ($\chi^2(1) = 11.60$, $p < 0.001$) rejects the null hypothesis of exogeneity, providing clear evidence that boarding congestion is endogenous and that the control-function correction is necessary. Importantly, even after accounting for endogeneity, the quadratic relationship between $Congestion$ and $TreatmentTime$ persists, with only modest changes in magnitude ($\beta_1 = 0.79$, $p < 0.001$ and $\beta_2 = -1.02$, $p < 0.001$).

To further assess whether the estimated nonlinear association between $Congestion$ and $TreatmentTime$ reflects an inverted U-shaped pattern or a monotonically increasing but concave

Table 1 Estimation Results

	First Stage		Baseline Model		Control Function	
	$Congestion$ (1)		$ln(TreatmentTime)$ (2)		$ln(TreatmentTime)$ (3)	
$NearestIP$	0.20***	(0.017)				
$Congestion$			0.65***	(0.159)	0.79***	(0.181)
$Congestion^2$			-1.04***	(0.228)	-1.02***	(0.209)
$WaitCensus$	0.07	(0.044)	-0.20	(0.195)	-0.20	(0.184)
$WaitCensus^2$	-0.11	(0.072)	0.45	(0.281)	0.44	(0.263)
$TreatmentCensus$	0.19	(0.107)	-0.24	(0.299)	-0.29	(0.295)
$TreatmentCensus^2$	-0.27*	(0.106)	0.26	(0.249)	0.41	(0.271)
$Workload$	-0.01	(0.006)	-0.02	(0.014)	-0.03	(0.014)
$Workload^2$	0.001**	(0.000)	0.00*	(0.000)	0.00*	(0.001)
$AvgTriage$	-0.07	(0.038)	0.03	(0.032)	0.03	(0.030)
$HospOccupancy$	0.75***	(0.131)	0.21***	(0.061)	0.08*	(0.033)
Residual $\hat{v}$					-0.85***	(0.249)
Other controls	Included		Included		Included	
Sample Size	470,173		470,173		470,173	
Adj-R^2	0.45		-		-	
Ancillary parameter σ	-		1.07		1.07	
Ancillary parameter κ	-		0.98***		0.98***	
Wald χ^2	-		135,112***		136,248***	

Notes. Clustered robust standard errors in parentheses. *$p<0.05$; **$p<0.01$; ***$p<0.001$.

relationship, we implement spline regressions (Tan and Netessine 2019, Kesavan et al. 2014). Specifically, we introduce specifications with one and two knots that partition boarding congestion into equally sized segments and estimate Equation 3 after replacing $Congestion$ and $Congestion^2$ with these segments. Across specifications, the coefficient on the first spline segment is positive and statistically significant ($\beta = 0.17$, $p = 0.026$ for one knot and $\beta = 0.41$, $p = 0.002$ for two knots), whereas the coefficient on the last spline segment is negative and significant ($\beta = -0.54$, $p < 0.001$ for one knot and $\beta = -0.77$, $p < 0.001$ for two knots). These findings indicate that treatment time initially rises with boarding congestion but declines beyond a threshold, providing clear evidence of an inverted U-shaped relationship.

To facilitate interpretation of the estimated coefficients, we plot the predicted treatment time as a function of boarding congestion for a representative hospital in Figure 2, with bars indicating 95% confidence intervals. The figure clearly illustrates an inverted U-shaped relationship between treatment time and boarding congestion[5]. For instance, as boarding congestion increases from 0.1 to 0.4–corresponding to an increase in boarding capacity utilization from 10% to 40%–the mean treatment time increases from 216 minutes to 257 minutes, a 19% increase. In contrast, as congestion

[5] The inverted U-shaped relationship is consistent across hospitals.

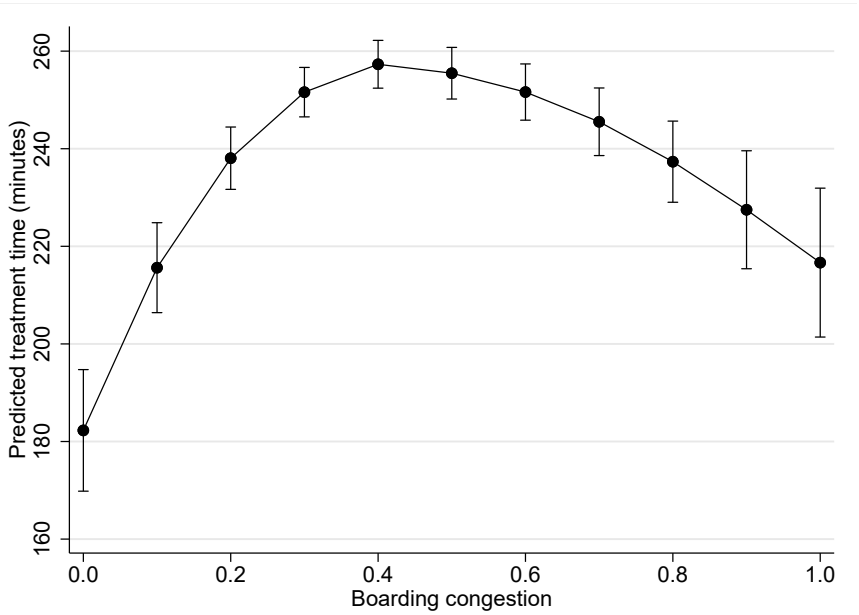


Figure 2 Effect of Boarding Congestion on Treatment Time

further increases from 0.4 to 0.7 (i.e., utilization rising from 40% to 70%), the mean treatment time declines from 257 to 245 minutes, representing a 4.7% decrease.

In summary, these results provide robust evidence that boarding congestion has a nonlinear effect on ED treatment time. The inverted U-shaped pattern suggests that moderate levels of boarding congestion exacerbate treatment delays, while extremely high congestion may trigger operational responses that partially mitigate these effects. This observation is noteworthy because it indicates that the consequences of boarding congestion extend beyond its well-documented adverse effects on the health outcomes of boarded patients. Instead, boarding congestion can also influence overall ED performance by altering treatment times for other patients who are not themselves boarded. As such, boarding congestion represents a system-level operational challenge with spillover effects across patient populations within the ED. We next examine the mechanism underlying our results.

4. Underlying Mechanism

We posit that the inverted U-shaped relationship between boarding congestion and ED treatment time is driven by the extent to which boarding patients continue to utilize ED resources during the boarding period (as illustrated in Figure 3). Building on prior literature and insights from clinicians at our partner hospitals, we first describe the theoretical mechanism underlying this relationship and then present empirical evidence in support of it.

Although the mechanism can, in principle, accommodate changes in congestion within a visit as the boarding environment evolves, congestion in our setting varies predominantly across visits

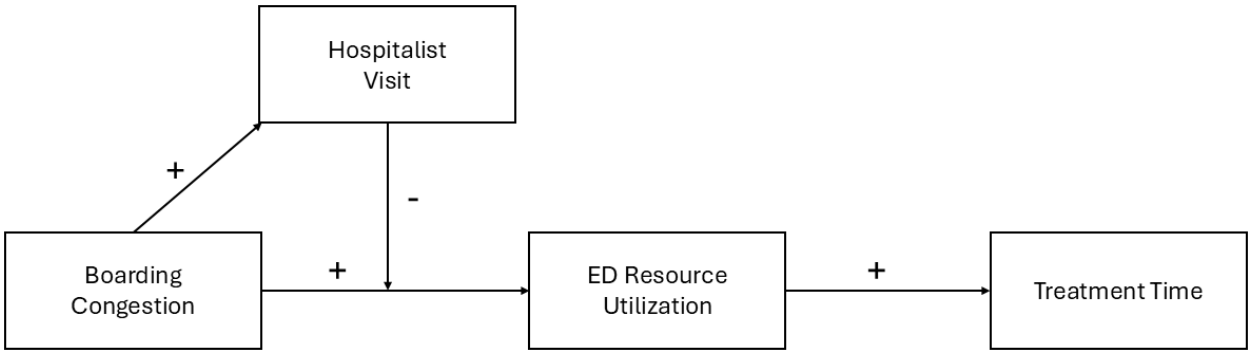


Figure 3 Theoretical model of the underlying mechanism

rather than within visits. We therefore develop the theoretical mechanism around differences in the overall congestion environment across visits.

4.1. Theory

Contrary to the common perception that boarding patients no longer rely on ED resources once they are transferred to the boarding area (Smalley et al. 2020), they may continue to receive care from ED staff, particularly when boarding durations are prolonged (Armony et al. 2015). This care can include routine checkups by ED staff as well as additional diagnostic tests ordered during the boarding period (Liu et al. 2011). For instance, in our sample, boarding patients remain in the boarding area on average for 15.2 hours, which is 2.16 times the average ED treatment time, with a 25th percentile of 2.42 hours, a median of 6.05 hours, and a 75th percentile of 21.9 hours. Moreover, 22% of boarding patients received at least one documented physician reassessment by an ED physician, while 9% had at least one diagnostic test ordered during boarding. Consequently, boarding patients may continue to draw on shared ED resources–such as physician and nursing time–even after the disposition decision has been made and they have entered the boarding stage. As boarding congestion increases, a larger number of boarding patients require such care during the boarding period, leading to greater utilization of ED resources. Because these resources are shared, higher utilization by boarding patients reduces the effective capacity available for ED patients undergoing treatment, resulting in delays and longer treatment times.

As discussed in Section 2.2, hospitalists are responsible for accepting boarding patients and coordinating their transfer to inpatient units, but they typically do not visit boarded patients immediately. Until a hospitalist arrives, ED staff retain responsibility for care. When a hospitalist visits, responsibility shifts from ED staff to the hospitalist. However, due to inpatient unit capacity constraints and administrative delays in bed assignment and transfers, boarding patients may remain physically in the boarding area for extended periods even after care responsibility has transitioned. As a result, some boarding patients initially receive care from ED staff and later from hospitalists, while

remaining physically in the boarding area. In our sample, boarding patients account for 16.1% of all ED visits. Among these patients, 67.1% are transferred to inpatient units without an observed hospitalist visit in the ED, while the remaining 32.9% receive care from ED staff early in the boarding period and from hospitalists thereafter.

We posit that hospitalist involvement negatively moderates the relationship between boarding congestion and ED resource utilization in two ways (Hrycko et al. 2019, Chadaga et al. 2012). First, once hospitalists assume care responsibility, boarding patients no longer require routine checkups from ED physicians, reducing demands on ED staff time (see Appendix A3 for the empirical evidence). Second, ED physicians are more likely than hospitalists to order diagnostic tests for boarding patients. This difference may stem from higher cognitive load and greater risk aversion among ED physicians (Soltani et al. 2022), as well as concerns that their decisions for boarding patients will later be reviewed by hospitalists (Deck and Jahedi 2015). In contrast, hospitalists typically have greater familiarity with patients' conditions and are more selective in ordering tests (Ma et al. 2005). Consequently, testing resources are less utilized when boarding patients are managed by hospitalists rather than ED staff (see Appendix A3 for the empirical evidence). Taken together, these arguments imply that the effect of boarding congestion on ED resource utilization is likely attenuated–and may even reverse–when care responsibility shifts from ED staff to hospitalists.

Boarding congestion also affects hospitalist behavior. In our partner hospitals, ED staff notify hospitalists of boarding patients awaiting inpatient beds, after which hospitalists independently decide when to visit based on competing inpatient responsibilities. As congestion in the boarding area increases, ED staff place more frequent requests, increasing the likelihood of hospitalist involvement.

Overall, this theory implies a nonmonotonic relationship between boarding congestion and ED treatment time. At low levels of congestion, boarding patients are primarily managed by ED staff, leading higher congestion to increase ED resource utilization and thus, ED treatment times. At higher levels of congestion, hospitalist involvement becomes more frequent, shifting care responsibility away from ED staff and reducing boarding patients' reliance on ED resources. This shift alleviates capacity constraints in the treatment area and shortens ED treatment times. In the next subsection, we present empirical evidence supporting this mechanism.

4.2. Empirical Evidence

Our theoretical mechanism implies that ED resource utilization mediates the relationship between boarding congestion and treatment time, while hospitalist visits moderate the relationship between boarding congestion and ED resource utilization. To empirically evaluate this mechanism, we specify the following mediation model as a recursive system of equations:

$$Pr(HospitalistVisit_{ih}=1) = Pr(\alpha_0 + \alpha_1 Congestion_{ih} + \alpha_2 Congestion_{ih}^2 + \mathbf{Controls'}_{ih}\boldsymbol{\alpha} + \nu_{ih} > 0)$$

$$\begin{aligned} EDUtilization_{ih} =& \lambda_0 + \lambda_1 Congestion_{ih} + \lambda_2 HospitalistVisit_{ih} \\ &+ \lambda_3 Congestion_{ih} \times HospitalistVisit_{ih} + \mathbf{Controls'}_{ih}\boldsymbol{\lambda} + \vartheta_{ih} \\ ln(TreatmentTime_{ih}) =& \beta_0 + \beta_1 EDUtilization_{ih} + \mathbf{Controls}_{ih}\boldsymbol{\beta} + \gamma\hat{v}_{ih} + \epsilon_{ih}, \end{aligned} \tag{4}$$

where (i) $HospitalistVisit_{ih}$ is an indicator equal to one if at least one hospitalist visit occurs during the treatment time of ED visit i at hospital h, (ii) $Pr(HospitalistVisit_{ih} = 1)$ denotes the corresponding probability, (iii) $EDUtilization_{ih}$ is the hourly average number of boarding patients utilizing ED resources–specifically, receiving a routine checkup from an ED physician or undergoing a diagnostic test at laboratory or imaging facilities–during the treatment time of ED visit i at hospital h, normalized by the number of licensed beds in each ED, (iv) $\mathbf{Controls'}_{ih}$ is a vector of ED- and hospital-level control variables along with seasonality and time-of-day fixed effects, and (v) ν_{ih} and ϑ_{ih} are random error terms. We estimate the $Pr(HospitalistVisit_{ih} = 1)$ equation using a probit specification.

Table 2 reports the results of the estimation. Within each column, the top block reports the estimates of the $Pr(HospitalistVisit_{ih} = 1)$ equation, the second block presents the estimates of the $EDUtilization$ equation, and the third block reports the estimates of the $ln(TreatmentTime_{ih})$ equation. Model-1 estimates Equation 4 including only main effects of $Congestion$ and $HospitalistVisit$ in the $EDUtilization$ equation. Model-2 adds the interaction between $Congestion$ and $HospitalistVisit$, corresponding to the full specification of Equation 4. Comparing Models 1 and 2 therefore allows us to assess the moderating effect of $HospitalistVisit$. Model-3 further augments the $ln(TreatmentTime_{ih})$ equation by including $Congestion$ and $Congestion^2$, enabling us to evaluate the extent to which the proposed mediation mechanism explains the inverted U-shaped relationship identified in Section 3.4.

Results from the $Pr(HospitalistVisit_{ih} = 1)$ equation show a nonlinear relationship between boarding congestion and hospitalist involvement. The linear term $Congestion$ is positive and statistically significant ($\alpha_1 = 2.18$, $p < 0.001$), while the squared term $Congestion^2$ is negative and statistically significant ($\alpha_2 = -1.83$, $p < 0.001$). Spline regressions with one and two knots–constructed as in Section 3.4–show that the coefficient on the first spline segment is positive and significant ($\beta = 1.03$, $p < 0.001$ for one knot and $\beta = 1.59$, $p < 0.001$ for two knots), whereas the coefficient on the last spline segment is not statistically significant ($\beta = -0.17$, $p = 0.117$ for one knot and $\beta = -0.03$, $p = 0.861$ for two knots). These findings indicate that hospitalist involvement is more likely at higher boarding congestion levels, supporting our theoretical prediction that care for boarding patients is primarily provided by ED staff at low congestion levels and is more likely to involve hospitalists at higher congestion levels.

Turning to the *EDUtilization* equation, adding the interaction between *Congestion* and *HospitalistVisit* in Model-2 improves model fit relative to Model-1 as evidenced by a lower BIC. The interaction term is negative and statistically significant ($\lambda_3 = -0.09$, $p < 0.001$), indicating that hospitalist involvement attenuates the effect of boarding congestion on ED resource utilization. In Model-2, the coefficient on *Congestion*–which captures the effect of boarding congestion when care is provided by ED staff (i.e., $HospitalistVisit = 0$)–is positive and significant ($\lambda_1 = 0.07$, $p < 0.001$). By contrast, the combined effect of *Congestion* and the interaction term ($\lambda_1 + \lambda_3 = -0.025$)–representing the effect when care responsibility is more likely shifted to hospitalists (i.e., $HospitalistVisit = 1$)–is negative and significantly smaller than the ED staff-led effect ($\chi^2 = 158.82$, $p < 0.001$). Together, these results align with the proposed mechanism that boarding congestion increases ED resource utilization when boarding patients are managed by ED staff, but reduces ED resource utilization when care responsibility is more likely assumed by hospitalists.

Finally, in the $ln(TreatmentTime_{ih})$ equation in Model-2, the estimated coefficient on *EDUtilization* is positive and significant ($\beta_1 = 2.22$, $p < 0.001$), indicating that greater utilization of ED resources by boarding patients increases treatment times for ED patients. Collectively, these findings provide empirical support for the presence of the mechanism illustrated in Figure 3.

The mediation analysis that leverages Model-3 shows that once hospitalist visits and ED resource utilization are included as a moderator and a mediator, respectively, the direct effects of boarding congestion and its squared term on ED patient treatment time in the $ln(TreatmentTime_{ih})$ equation are no longer statistically significant ($\beta = -0.01$, $p = 0.965$ for *Congestion* and $\beta = -0.32$, $p = 0.207$ for $Congestion^2$). In contrast, the indirect effects operating through ED resource utilization are significant ($\beta = 0.05$, $p = 0.002$ for *Congestion* and $\beta = -0.05$, $p = 0.008$ for $Congestion^2$). This pattern provides evidence that the proposed mechanism illustrated in Figure 3 serves as a key mediator of the inverted U-shaped relationship between boarding congestion and ED treatment time.

In summary, at low levels of boarding congestion, ED staff provide most of the care for boarding patients, leading congestion to increase ED resource utilization and treatment times. This mechanism explains the upward-sloping portion of the inverted U-shaped relationship. At higher levels of boarding congestion, hospitalist involvement is more likely, shifting care responsibility away from ED staff and reducing boarding patients' reliance on ED resources. This shift alleviates capacity constraints in the treatment area and shortens treatment times, generating the downward-sloping portion of the inverted U-shaped relationship observed in our empirical analysis. As demonstrated in Appendix A2, we conduct an extensive set of robustness tests and find that our main results and the underlying mechanism are robust to several alternative explanations and model specifications.

Table 2 Mediation Analysis Results

	Model-1	Model-2	Model-3
	$HospitalistVisit$	$HospitalistVisit$	$HospitalistVisit$
$Congestion$	2.18***	2.18***	2.18***
	(0.213)	(0.213)	(0.213)
$Congestion^2$	-1.83***	-1.83***	-1.83***
	(0.244)	(0.244)	(0.244)
$Controls'$	Yes	Yes	Yes
	$EDUtilization$	$EDUtilization$	$EDUtilization$
$Congestion$	0.02	0.07***	0.07***
	(0.022)	(0.021)	(0.021)
$HospitalistVisit$	-0.01**	0.03***	0.03***
	(0.002)	(0.003)	(0.003)
$Congestion \times HospitalistVisit$		-0.09***	-0.09***
		(0.007)	(0.007)
$Controls'$	Yes	Yes	Yes
	$ln(TreatmentTime)$	$ln(TreatmentTime)$	$ln(TreatmentTime)$
$EDUtilization$	2.22***	2.22***	2.15***
	(0.492)	(0.492)	(0.525)
$Congestion$			-0.01
			(0.238)
$Congestion^2$			-0.32
			(0.250)
$Controls$	Yes	Yes	Yes
Sample size (N)	470,173	470,173	470,173
BIC	654,830.5	642,520.2	642,439.4

Notes: Clustered robust standard errors in parentheses. * $p<0.05$, ** $p<0.01$, *** $p<0.001$.

The underlying mechanism highlights the management of ED resource utilization by boarding patients and the timing of hospitalist visits as key operational levers for mitigating the adverse effects of boarding congestion. We operationalize these strategies in the next section.

5. Operational Interventions

In this section, we begin by constructing a simulation model that captures key operational features of real-world EDs. We then proceed to calibrate the input parameters of the simulation model using data from one of our tertiary EDs. Next, we introduce two easy-to-implement interventions: (i) a threshold-based patient-prioritization policy for allocating ED resources between treatment patients and boarding patients requiring follow-up care, and (ii) hospitalist visit scheduling policies for coordinating hospitalist visits to boarding patients in the ED. The policy parameters are determined numerically through simulation-based neighborhood search. We then conduct a sensitivity analysis on various ED characteristics, such as patient arrival patterns and the complexity of patients treated in the ED, to provide insights into which types of EDs can benefit more from the suggested policies. Finally, we examine the joint implementation of the two easy-to-implement policies.

5.1. Simulation Design

Following prior studies (e.g., Saghafian et al. (2012), Gomez et al. (2024)), we simulate the ED using a discrete-event simulation. We randomly select one of the tertiary EDs in the study and calibrate parameters based on the data from this ED. These parameters can be easily adjusted for other study EDs, as the patient flow is similar across all study EDs.

5.1.1. Simulation Model Following prior studies (e.g., Saghafian et al. (2012), Gomez et al. (2024), Feizi and Baker (2021)), we simulate the ED using discrete-event simulation, and simulation parameters are calibrated based on our empirical findings and observed data. The four main components of our simulation model are: (1) patient arrival, (2) patient classes, (3) services, and (4) hospitalist visits, which are explained below.

Patient Arrival: Following methodologies in previous studies (e.g., Saghafian et al. (2012), Gomez et al. (2024), Kim et al. (2015), Li et al. (2023), Chan et al. (2021)), we employ a Poisson process with time-varying rates to model arrivals. Similar to the methodology proposed by Gomez et al. (2024), empirical arrival rates are estimated as the total number of individuals arriving within a specific one-hour window on a given day of the week (e.g., from 1 a.m. to 2 a.m. on Monday), divided by the number of times this time window is observed in the data. The arrival rate $\lambda(t)$ at time t is then considered to be the empirical estimate for the given hour and day of the week corresponding to t. We note that these empirical estimates are maximum likelihood estimators under the assumption that the counts of arrivals over the produced windows are independent and identically distributed Poisson variables, which has been shown to be a reasonable assumption (Kim et al. 2015, Gomez et al. 2024).

Patient classes: Following Li et al. (2023) and Saghafian et al. (2012), we categorize patients into three classes instead of considering all five triage levels. This decision aligns with our goal of evaluating the impact of boarding congestion on the treatment time of patients in the treatment section (primarily patients with triage level 2-4) independently of their triage level. Class 1 comprises patients with CTAS 1, having the highest priority and usually receiving immediate treatment. Class 2 includes patients with CTAS levels 2–4, who may need to wait for treatment if all physicians are occupied. Finally, class 3 consists of patients with CTAS 5, having the least-urgent conditions. They are often tracked separately because their treatment needs are relatively simpler, and for more than 98% of cases, they have only a single interaction with the ED physician. Note that the patient's profile (including the number of labs and tests, required physician visits, service time in different treatment phases, disposition, checkup request during boarding, boarding time, etc.) of a newly generated patient in our simulation model is randomly sampled from the dataset of patients who arrived at the ED during that hour on the particular day of the week (bootstrap).

Services: As described in Section 3.1, the patients go through up to five phases of service in the ED. These phases are: 1) triage, 2) consultation with ED physicians for tests or medication orders (consultation), 3) processing of orders (diagnostics), 4) treatment and monitoring, and 5) boarding for admitted patients. Patients in the boarding phase may require ongoing monitoring, further consultations with ED physicians, and additional tests. Phases 2 to 4 can be repeated for a patient, indicating multiple instances of test orders and visits to ED physicians during their treatment, with the details determined by the patient's profile.

Upon arrival, patients are assessed and categorized according to their condition, known as triage codes, in Phase 1. After triage, they wait in the waiting area for an ED physician to become available. When an ED bed or treatment room opens up, a patient from the waiting area is selected to move to the ED bed/treatment room. The selection of the next new patient for assigning a bed/treatment room is based on the standard ED protocol used across the study EDs. The number of ED beds is set according to each ED's capacity.

The patients who leave without being seen (LWBS) are not explicitly represented in the simulation model. While our dataset identifies this group of patients through the discharge disposition field, it does not record the timing of abandonment. Explicitly modelling LWBS behavior would therefore require additional assumptions regarding abandonment timing and its relationship to waiting conditions, which are not directly supported by the data and could introduce model-driven bias. Importantly, LWBS cases represent less than 5% of all ED visits in our dataset and are predominantly low-acuity (CTAS 5) patients, who are less likely to require substantial treatment resources or contribute to downstream congestion and boarding processes. Moreover, because LWBS patients do not enter the treatment section, they do not contribute to treatment–section congestion – the main outcome of our policy evaluation – or to boarding workload. In practice, LWBS would only reduce the realized arrival rate to treatment. Since simulation arrival rates are calibrated from observed arrivals following Gomez et al. (2024), the empirical flow driving downstream treatment and boarding dynamics is preserved. Modeling LWBS would therefore mainly affect waiting-room occupancy, which is outside the scope of our policy comparison.

During Phase 2, the patient is evaluated by an ED physician. The number of ED physicians on duty at any given time, denoted as $m_2(t)$, varies according to the ED shift schedule. After the initial assessment, some patients may leave the ED, while others may be asked to undergo diagnostic tests such as blood work or imaging. Patients requiring diagnostics enter the testing queue, complete the tests, and await results in Phase 3.

In Phase 4, patients with available test results are attended to by the ED physicians responsible for orders in Phase 2. Each ED physician can manage up to s_{2-4} patients in the treatment section (i.e., patients in Phases 2-4). ED physicians work independently and may be responsible for several

patients at a time. In addition, as an ED physician rarely moves to another patient in the middle of their current interaction, we assume a non-preemptive process for provider-patient interactions. This non-preemptive and non-collaborative interaction aligns with our observations in the EDs we studied. Physicians typically avoid preemptive policies because they are inefficient, requiring the physician to re-review patient history and condition upon their return, and they can irritate patients.

After the evaluation in Phase 4, the patient is either admitted to the hospital or discharged. If admitted, the patient remains in the ED bed until transfer to the hospital ward (Phase 5). If discharged, the bed becomes available for other patients. Boarding patients may require follow-up visits from an ED physician as well as additional tests or laboratory work (checkup requests) until they are transferred to an inpatient bed. The timing of these checkup requests is determined by the patient's initial profile at the beginning of the simulation. In addition, observed boarding time may reflect not only inpatient capacity constraints and hospitalist availability, but also downstream logistical frictions such as transport staff availability, elevator delays, bed turnover, and transfer coordination. Because these components are not separately observed in our data, they are incorporated in reduced form through the empirical boarding-time distributions used to calibrate the simulation.

After completing an interaction with a patient, the ED physician's next visit is determined by the patient flow control policy. In the current practice of the studied EDs, priority is given to boarding patients who require the ED physician's attention. If multiple boarding patients are waiting to be seen by ED physicians, they are visited based on a first-come, first-served (FCFS) rule, considering their checkup request time. If no boarding patients are waiting for attention, ED physicians prioritize patients in Phase 4 (treatment) over those in Phase 2 (ordering). Additionally, if a physician has fewer patients than the upper bound they are willing to handle (i.e., s_{2-4}), they may choose to begin a new interaction by seeing a patient who has been assigned to a bed but is waiting for a physician to become available. Similarly, in Phase 3, according to the current practice, priority is given to boarding patients over treatment patients.

Hospitalist visits: The study EDs do not follow a formal protocol for hospitalist visits to boarding patients. Instead, the timing of these visits varies with ED operating conditions and the level of boarding demand. To capture this feature, we model hospitalist visits in the simulation as an endogenous hourly process that depends on boarding congestion, waiting-room congestion, treatment-section congestion, and recent hospitalist visit history. This formulation reflects the fact that hospitalist visits are responsive to system conditions and are not independent over time. We therefore use a probit-based decision rule, rather than a Poisson arrival process, to determine whether a hospitalist visit occurs in each simulated hour.

At each simulated hour, the model updates the ED state and determines whether a hospitalist visit occurs. When a hospitalist visit takes place, eligible boarding patients who would otherwise require follow-up attention from ED physicians during the boarding phase are instead seen by the hospitalist. As a result, hospitalist visits reduce the demand placed on ED physician capacity and help alleviate the operational burden created by boarding patients. Figure 4 presents the flowchart of the simulation model.

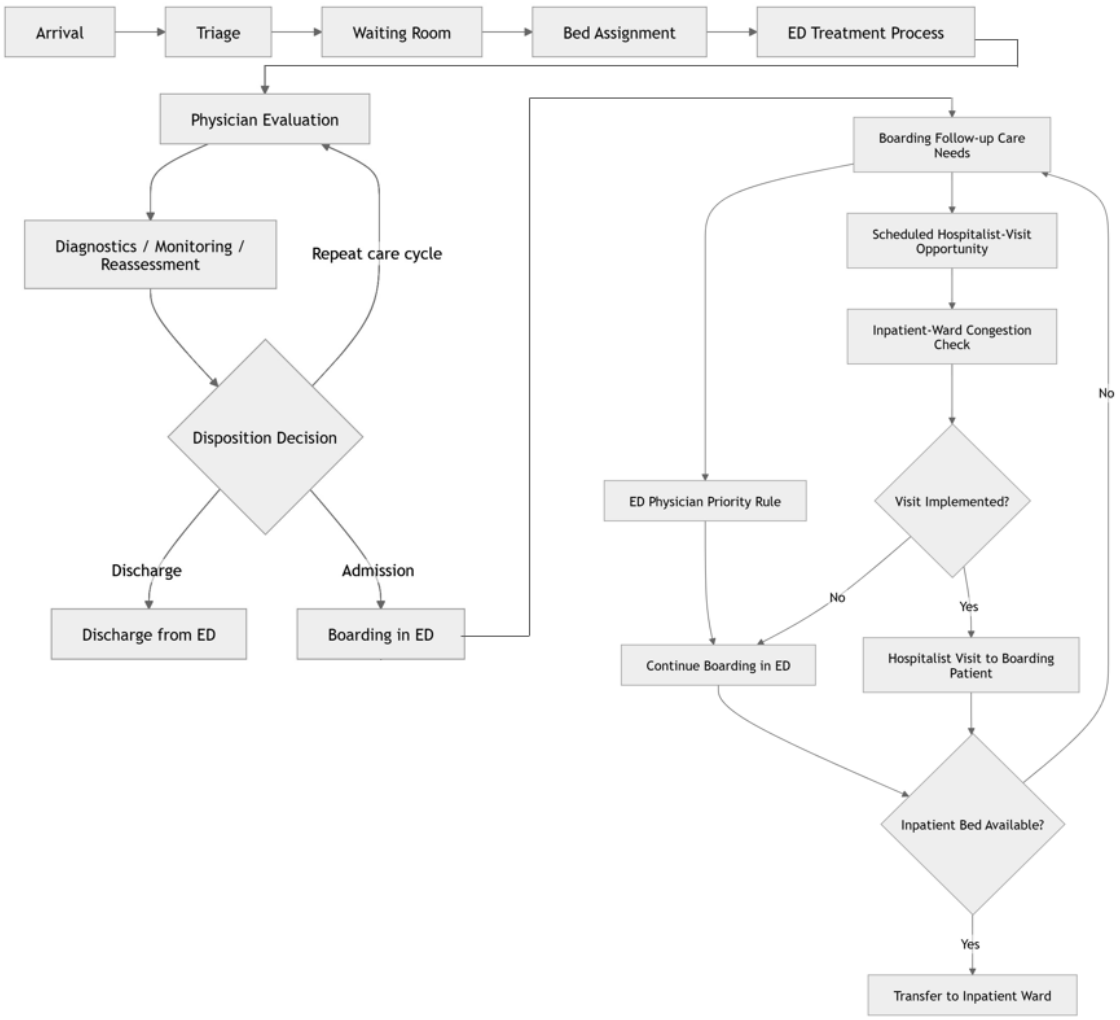


Figure 4 **Overview of the Simulation Model** ***Note:*** **The figure shows the full simulation framework, including hospitalist-visit scheduling (HVS) and the treatment-section patient flow control (threshold) policy. The baseline simulation follows the same core ED flow without activating these intervention-specific rules.**

Calibration of the Simulation Input Parameters: Parameters are primarily obtained from data directly, or, if that is not feasible, through matching simulation outputs with data estimates. Certain parameters cannot be directly derived from data and must be calibrated through simulation;

notably, this includes the service time in Phase 3 and the hospitalist-patient service time. These service times cannot be directly determined from data, as the recorded times only reflect event occurrences and the duration between events, which also accounts for the waiting time of a patient during a specific service phase.

To estimate these unknown parameters, we employ the approach proposed by Gomez et al. (2024). From the data, empirical estimates for five key quantities are derived: the time taken to be assigned to a bed (rooming), the time until the first order, the time to disposition, boarding time, and the total treatment duration $\hat{y}$. Unknown parameters are represented by vector x. For each selected option in x, the simulation model provides a vector $y(x)$. A nonlinear optimization process (surrogate optimization) is then conducted to search for x that minimizes the objective function: $||y(x) - \hat{y}||$, representing the Euclidean norm between the vector of empirically-estimated quantities and the vector of simulation-estimated quantities. Final parameters used for simulation are reported in Table 3.

Table 3 Simulation Parameters

Type	Description	Symbol	Value
Service Rates (Persons per Hour)	Processing of Orders	μ_3	3.4
	Processing of Boarding Patients with ED Physician	$\mu_{b,2}$	1.4
	Processing of Boarding Patients with Hospitalist	μ_h	0.9
Workers and Beds	Processing of Orders	m_3	15
	Beds	n	46
	ED Physician Capacity	s_{2-4}	8

Note that this table includes parameters calibrated using simulation and fixed variables. Other variables, including the number of orders per patient, patient arrival rate, and patient-physician service time in Phases 2 and 4, are sampled directly from the data.

Validation and Simulation Runs: We validate our simulation model for accuracy using established practices in prior literature (e.g., Sargent (2010), Lee et al. (2015), Feizi et al. (2023)). This validation process involves comparing the simulation model with the actual "real-world" ED system, focusing on key system features relevant to our study. These features include ED treatment time, boarding census, treatment section census, waiting room census, and waiting time[6]. As depicted in Table 4, the simulation results closely align with the existing ED system, indicating the accuracy of the model.

We follow a similar approach to prior studies (Saghafian et al. 2012), by choosing the number of replications to ensure that the confidence intervals are tight enough for reliable sample averages.

[6] It is important to note that there is no overlap between these system features and simulation input parameters. For instance, ED treatment time is different from the service time parameters for Phases 2-4, which are used as input parameters in the simulation. Treatment time is defined as the duration from ED bed assignment to disposition, including both service time and waiting time in Phases 2-4. Similarly, the waiting time in the waiting room before moving to the treatment section depends on ED physician and bed availability.

Therefore, there is no need to visually represent these very tight intervals. As a result, we do not explicitly report the confidence intervals in our presentation of the simulation results. All numerical results are the average of 500 simulation runs, each spanning a 24-week time frame with a two-week warm-up period.

Table 4 Comparison between actual and simulated ED.

	Waiting census	Waiting time (hr)	Treatment census	Treatment time (hr)	Boarding census
ED statistics	15.17	5.85	35.07	6.21	21.65
Simulation values	15.23	5.91	34.91	6.25	22.04
p-value	0.225	0.183	0.192	0.263	0.093

It is worth mentioning that the number of people in the waiting room, treatment section, and boarding section of the ED is not directly set as an input in the simulation model. Instead, it is determined by the simulated patient flow dynamics within the model. This approach helps to accurately represent the complex interactions and dynamics observed in real-world ED systems.

5.2. Patient Flow Control in Treatment Section

The findings in Section 4 reveal that boarding patients' use of ED resources during boarding increases treatment time for patients already in the treatment section. This highlights the importance of managing the return of boarding patients to the treatment section and prioritizing between boarding patients and those already undergoing treatment.

In this section, we propose a heuristic policy on prioritizing between boarding patients who need ED resources and those already undergoing treatment. The policy is inspired by previous policies proposed in the literature for patient flow controls in healthcare systems such as hospitals (e.g., Helm et al. (2011)). This policy, referred to as threshold policy (TP) throughout the paper, prioritizes treatment patients over boarding patients under specific conditions to ensure a smoother patient flow.

The TP policy draws on an operational approach already common in EDs. Boarding patients sometimes require follow-up evaluation by an ED physician while awaiting transfer to an inpatient unit. Consequently, after completing a task, an ED physician may face two simultaneous demands: a patient currently undergoing treatment and a boarding patient with a pending checkup request. Based on our observations and discussions with clinicians, the predominant practice in the study EDs is to prioritize the boarding-patient request. Importantly, TP does not propose that ED physicians begin prioritizing boarding patients. Instead, it asks whether a state-dependent prioritization rule should replace the existing unconditional boarding-first rule. The operational rationale is that both patient classes compete for the same limited physician resource. Giving priority to a boarding patient

for a checkup delays the treatment queue, whereas delaying the checkup increases the boarding checkup queue and its waiting time. Since neither class can be ignored forever, the optimal priority depends on the congestion levels of both queues.

Under TP, boarding patients requiring ED physician attention enter the checkup queue, while patients undergoing active treatment form the treatment queue. Treatment patients receive priority when treatment-section congestion exceeds a threshold. At intermediate levels of treatment congestion, priority also depends on the size of the checkup queue: treatment patients are prioritized when the checkup queue remains below a threshold, whereas boarding patients regain priority as their pending-care queue becomes sufficiently large. Thus, TP protects treatment throughput during periods of substantial treatment-section congestion while preventing excessive delays in boarding-patient follow-up care.

We evaluate TP's effectiveness by comparing it with two benchmark scenarios. The first scenario is the current practice, where boarding patients are prioritized over emergency patients for ED resources; We therefore model this boarding-first rule as the status quo. The second scenario assumes that boarding patients are redirected to a separate system, thereby removing any additional burden on ED resources. This scenario represents a lower bound for treatment time and is referred to as the "lower bound". By comparing TP with these benchmarks, we assess its performance and potential for improvement. We use two metrics for comparison: average treatment time and treatment section census.

In TP, boarding patients requiring ED resources enter a queue called the checkup queue, while patients currently undergoing treatment in the treatment section are referred to as treatment patients. Treatment patients are prioritized over those in the checkup queue when either the number of treatment patients exceeds a threshold (L_1) or falls between L_1 and L_2, provided that the checkup queue is below another threshold (L_3).

Here, L_1 and L_2 serve different purposes. L_1 captures situations in which treatment-section congestion is sufficiently high that treatment patients should be prioritized regardless of the size of the checkup queue. In contrast, L_2, together with L_3, defines an intermediate operating region in which prioritization depends jointly on treatment–section congestion and the buildup of the boarding checkup queue. This structure allows the policy to balance the needs of patients currently in treatment against the waiting burden of boarding patients more flexibly than a single-threshold rule.

Drawing on previous studies (e.g., Helm et al. (2011)), we numerically calculate these threshold values using simulation-based neighborhood search, optimizing for average treatment and waiting times of patients in the checkup queue as our objective function. We expect that these thresholds are influenced by the total number of patients in the boarding section, as the checkup queue's status

depends on the boarding patient count. Accordingly, we derive L-values for low, moderate, and high boarding census levels.

We begin by identifying the first and third quartiles of the boarding census distribution. We then determine the optimal values of L_1, L_2, and L_3 for three scenarios: when the boarding census is below the first quartile (subscript l), between the first and third quartiles (subscript m), and above the third quartile (subscript h). This approach allows us to adjust treatment priority based on varying boarding census levels, providing a more adaptable and practical solution. We expect that as the boarding census increases, prioritizing patients in the checkup queue becomes more critical, leading to the relationship $L_1^h \geq L_1^m \geq L_1^l$ and $L_2^h \geq L_2^m \geq L_2^l$. Additionally, we ensure that $L_2 \leq L_1$.

Results: The results for the selected ED are summarized in Table 5. The first row of the table displays the outcomes obtained under the current policy, which aligns well with the observed data. Comparison between the status quo and the lower bound reveals that boarding patients can increase the treatment time by approximately 12%, highlighting the need for ED administrators to consider the additional workload caused by boarding patients when making operational decisions, such as staffing levels.

Our analysis strongly supports the effectiveness of the proposed TP in mitigating the negative impact of boarding congestion on treatment time. Specifically, with TP implementation, we observe an average reduction of 2.23 patients in the treatment section (corresponding to a 37.48% improvement in treatment–section congestion) and an average decrease of 1.27 patient-years in total treatment time (representing a 26.73% improvement) compared to the current policy.

Table 5 Simulation Results of Boarding Patient Flow Control Policies

Policy	Treatment time (hr)	Hourly average treatment census	Total treatment time* (patient-years)
Status quo (SQ)	6.25 (6.23, 6.27)	34.91 (34.58, 35.24)	44.28 (44.18, 44.38)
Lower Bound (LB)	5.58 (5.57, 5.59)	28.96 (28.81, 29.11)	39.53 (39.45, 39.61)
Threshold Policy (TP)	6.07 (6.05, 6.09)	32.68 (32.43, 32.93)	43.01 (42.93, 43.09)
Δ (%)	26.86%	37.48%	26.73%

Notes. Pct. improvement compared to the status quo and lower bound presented in the last row (i.e., (SQ-TP)/(SQ-LB)·100). Confidence intervals in parentheses for 500 iterations. * Total treatment time is calculated as (average treatment time · total annual patient volume)/(24 · 365)

If the optimal value of L_1 is set to 1, treatment patients will always be prioritized over boarding patients. Conversely, if L_3 is set to 1 and L_1 is set to a very high value, this indicates that, under the optimal policy, boarding patients will always take precedence over treatment patients (i.e., the

status quo would be optimal). Two factors influence the optimal values of L_1, L_2, and L_3: (1) the objective function and (2) the characteristics of the ED.

We determined the optimal threshold values (i.e., L_1, L_2, and L_3) based on the combined total of average treatment and waiting times, as advised by our partner hospitals. However, in certain emergency departments (EDs), a one-minute increase in treatment time does not necessarily correspond to a one-minute increase in waiting time for patients in the checkup queue. Consequently, EDs can refine their objective by using a weighted average of treatment time for patients receiving care and waiting time for those in the checkup queue, thereby reflecting the differing importance of wait times for treatment versus boarding patients. When more weight is placed on treatment patients, the optimal policy favors prioritizing them over boarding patients more often than when greater weight is assigned to boarding patients.

To understand how ED characteristics influence the optimal prioritization policy, we focus on five characteristics that directly determine the relative demand and service burden generated by treatment and boarding patients in our simulation: (1) the arrival rate of ED patients, (2) the arrival rate of checkup requests from boarding patients, (3) the probability that an ED patient is admitted and subsequently enters boarding, (4) the service requirement of a boarding-patient checkup, and (5) the service requirement of treatment patients. These characteristics were selected because together they capture the principal demand-side and service-side drivers of the two patient queues competing for ED physician capacity. To evaluate the impact of these characteristics on the optimal policy, we re-ran the simulation model by changing each of these characteristics by ±25% (see Table A.11 in Appendix A4). We summarize our findings in Figure 5.

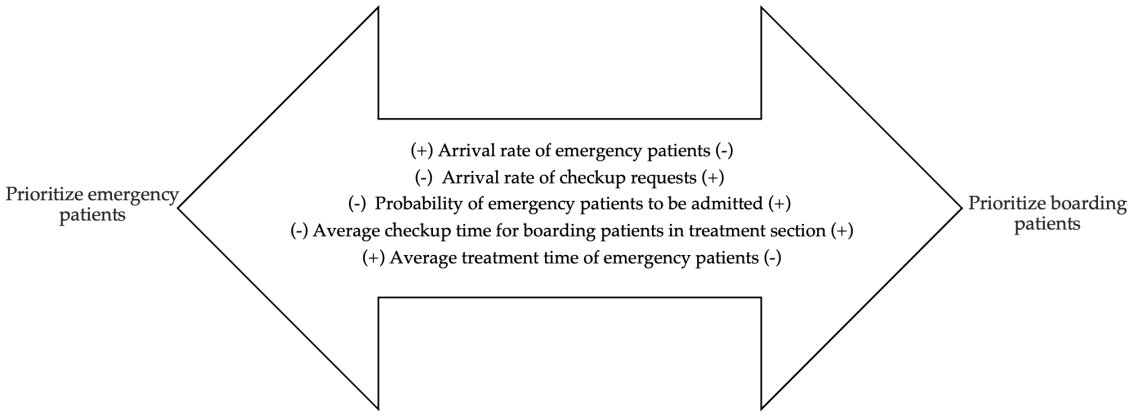


Figure 5 Patient Flow Control Based on Key Characteristics of the ED

The direction of each relationship in Figure 5 follows from its effect on the relative workload of the two queues. First, a higher arrival rate of ED patients increases the inflow to the treatment section

and causes treatment-patient congestion to accumulate more rapidly; consequently, the policy shifts toward prioritizing treatment patients. Conversely, when the treatment-patient arrival rate is lower, more physician capacity can be allocated to boarding-patient follow-up.

Second, a higher arrival rate of checkup requests increases the recurring workload generated by patients already in boarding. If these requests are deferred, the boarding checkup queue accumulates and continues competing for physician capacity. The optimal policy therefore shifts toward boarding patients as the checkup-request rate increases, whereas a lower request rate favors treatment patients.

Third, a higher admission probability increases the fraction of treatment patients who subsequently enter boarding and may generate follow-up demand. This enlarges the downstream boarding workload and increases the benefit of preventing a buildup of unattended boarding patients; therefore, higher admission probabilities shift priority toward boarding patients. When admission probability is lower, fewer treatment patients become boarders and the policy places relatively greater priority on treatment patients.

Fourth, longer service times for boarding-patient checkups make each boarding request more resource intensive and reduce the rate at which the boarding checkup queue can be cleared. The resulting accumulation increases the future burden imposed by the boarding queue, causing the optimal policy to place greater priority on boarding patients. Shorter boarding checkups have the opposite effect because their queue can be cleared more readily.

Finally, longer treatment times make treatment patients consume physician capacity for longer periods and cause congestion in the treatment section to accumulate more rapidly. The optimal policy therefore shifts toward treatment patients when their treatment requirements are longer; when treatment patients can be processed more quickly, relatively more capacity can be devoted to boarding-patient follow-up.

From the perspective of the hospital administrator, these analyzes demonstrate that prioritizing boarding patients is best suited for EDs with (i) a high influx of checkup requests from boarding patients, (ii) a high likelihood of emergency patients being admitted to the hospital, and (iii) longer checkup times for boarding patients in the treatment section. These conditions are commonly observed in EDs that handle a more acute and/or complex patient mix, including patients with lower triage codes, more comorbidities, or older ages. As a result, trauma centers or teaching hospital EDs, which often manage such patient mixes, can benefit from prioritizing boarding patients over treatment patients. Conversely, EDs managing less acute or severe cases, such as community hospitals, might find it more advantageous to prioritize treatment patients over boarding patients.

Implementing a prioritization policy requires monitoring separate queues for treatment and boarding patients. One approach involves tracking the real-time number of patients in each queue and

adjusting the prioritization scheme accordingly. This would require a queue management system (QMS) to provide ED staff with real-time status updates, enabling efficient patient flow management without extensive modifications to existing infrastructure or procedures.

5.3. Hospitalist Visit Scheduling (HVS)

In Section 4, we show that hospitalist visits can mitigate the burden that boarding patients impose on ED resources. Building on this insight, this section develops and evaluates hospitalist visit scheduling (HVS) policies for the boarding section. Importantly, the operational feasibility of these policies depends not only on boarding demand in the ED but also on congestion in the corresponding wards. When wards are highly occupied, hospitalists may have limited flexibility to make additional visits to the ED, implying that scheduled visits do not always translate into actual visits. We therefore extend the simulation to account for inpatient congestion and use it to evaluate two classes of HVS policies—census-based and time-based scheduling—thereby providing a more realistic assessment of both their effectiveness and their feasibility (for related insights on scheduling policies in healthcare settings, see, e.g., Robinson and Chen (2003, 2010)).

One challenge in designing HVS policies is the heterogeneity across inpatient services and hospitalists. The number of boarding patients requiring hospitalist attention can vary substantially across wards, depending on specialty and patient mix. In addition, hospitalists differ in their scheduling preferences and inpatient responsibilities. Because hospitalists must also manage rounds and other ward-based activities, visits to boarding patients in the ED may be more difficult to accommodate when the corresponding ward is highly occupied. As a result, the feasibility of a hospitalist visit depends not only on boarding demand in the ED but also on the level of congestion and operational pressure in the inpatient services. Moreover, some hospitalists may prefer fixed visit times over irregular visits triggered by fluctuations in the boarding census (Zhu 2018).

To capture these operational realities, we evaluate potential HVS policies that schedule hospitalist visits based either on the current boarding census or on predetermined time intervals. According to the census-based policy, a hospitalist visit is scheduled once the number of boarding patients awaiting admission to the corresponding ward reaches a specified threshold, called the census-based scheduling threshold. Under the time-based policy, hospitalist visits are scheduled at regular predetermined intervals (e.g., every 6 hours), regardless of the current boarding census; this interval is referred to as the time-based scheduling threshold. However, in both cases, a scheduled visit does not automatically translate into an actual visit. Rather, the implementation of the visit depends on the availability of the hospitalist, which is influenced by congestion in the corresponding ward.

To account for inpatient-side congestion, we augment the simulation so that a scheduled hospitalist visit is not always implemented. For the ward associated with each hospitalist, we define congestion

states by using occupancy percentiles, specifically the 75th and 90th percentiles observed in the data. This method allows the model to identify congestion patterns unique to our system, rather than relying on fixed thresholds. This approach aligns with existing literature in healthcare operations, which indicates that performance tends to deteriorate when occupancy levels become too high (Forster et al. 2003, Pratt and Wood 2021). Conditional on congestion states, we model the feasibility of scheduled hospitalist visits using a parsimonious, monotone probability structure (1, 0.75, and 0.5 for low, moderate, and high inpatient-side congestion, respectively)[7], which serves as a reduced-form approximation of decreasing operational flexibility as inpatient congestion intensifies. These probabilities are not intended to represent structural estimates, but rather calibrated parameters that capture operational frictions in a tractable manner, consistent with prior simulation-based approaches in healthcare operations (Saghafian et al. 2012, 2015, Helm et al. 2011, Gunal and Pidd 2010). This approach enables us to incorporate operational constraints on the inpatient-side while maintaining tractability and avoiding additional assumptions that cannot be directly identified from the data. We verify that our qualitative insights are robust to alternative specifications through sensitivity analysis.

To capture inpatient-side congestion, we extend the simulation to model the occupancy dynamics of the destination wards explicitly. Ward occupancy is driven by two sources of inflow: admissions from the ED boarding process and non-ED admissions, such as scheduled surgical admissions and other direct inpatient admissions. For the latter group, both the arrival rate and the length of stay are determined through the calibration procedure described in Section 5.1.1, so that the simulated level of inpatient congestion is aligned with the patterns observed in the data. For ED patients admitted to the hospital, we observe their inpatient length of stay in the data and incorporate it into the simulation through random sampling when generating each patient profile, as discussed in Section 5.1.1. As a result, ward occupancy evolves endogenously over time based on both admitted ED patients and other inpatient admissions.

We use a greedy search to determine the optimal levels for the census-based and time-based scheduling thresholds for their respective policies. For each threshold value, we run the simulation model to estimate expected performance based on the defined objective function. The optimal threshold is then selected as the one that achieves the lowest objective function value. Our objective function considers two main goals: (i) to minimize the additional workload on the treatment section due to boarding patients by maximizing the number of boarding patients seen by a hospitalist in the early hours of boarding time, prior to a checkup request, and (ii) to minimize the frequency of

[7] We later conduct sensitivity analyses using alternative probability values to assess the robustness of the results to different assumptions regarding the effect of inpatient congestion on the feasibility of scheduled hospitalist visits.

hospitalist visits. To this end, we introduce a new performance metric alongside those discussed in the previous section: the daily average number of hospitalist visits.

To accommodate varying patient volumes across wards, we establish an optimal scheduling threshold for each policy based on different demand rates. Our simulation model incorporates a demand rate by assuming that a specific percentage of boarding patients can be visited by a specific hospitalist during their boarding phase. For example, with a demand rate of 0.4, we estimate that 40% of patients generated in the simulation can be attended to by the hospitalist while in boarding. This detail is embedded in each patient's profile upon generation in the simulation model.

This approach enables hospitalists to select the most suitable policy—whether time-based or census-based scheduling—by taking into account both their preferences and the specific demand in their inpatient services. This framework aids in policy selection and streamlines the decision-making process.

Results: Our proposed framework is summarized in Table 6. With this framework, EDs can seamlessly choose the optimal policy by taking into account the demand rate and hospitalists' preferences regarding time- and census-based scheduling.

Table 6 Scheduling Framework for a Hospitalist Based on Demand Rates

	Demand rates									
Policy	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Census-based (number of patients in boarding*)	2	3	3	4	5	6	6	7	8	8
Time-based (hr)	12	8	8	6	6	4	4	4	3	3

*: Number of boarding patients who will be admitted to the hospitalist's wards

Table 7 shows the results for the status quo and the threshold policy (TP).[8]" A comparison between time-based and census-based policies reveals that, despite the preference of hospitalists for fixed visit times in the time-based policy, the census-based policy yields fewer hospitalist trips to the boarding section and outperforms the time-based policy with respect to average treatment time and census.

Implementing this operational policy can reduce the adverse impact of boarding patients on treatment–section congestion by 31.4%, corresponding to an average reduction of 1.87 patients in the treatment section. When both interventions are implemented together, the adverse impact of boarding patients on treatment–section congestion can be reduced by 44.37%, corresponding to 2.64 fewer patients in the treatment section on average, and treatment time improves from 6.25 hours to 5.98 hours (approximately 16 minutes per patient). Importantly, these improvements are achieved with only a 7% increase in the average daily number of hospitalist visits to the ED, from 2.97 to 3.18, which corresponds to an increase of only 0.21 visits per day on average.

[8] The threshold policy refers to the patient flow control intervention discussed before.

Table 7 Simulation Results of Hospitalist Visit Scheduling (HVS) Policies

	Average treatment time (hr)		Hourly average treatment section census		Daily average number of hospitalist visit per specialty*	
Policy	Status quo	TP	Status quo	TP	Status quo	TP
Without HVS**	6.25 (6.23, 6.27)	6.07 (6.05, 6.09)	34.91 (34.58, 35.24)	32.68 (32.43, 32.93)	2.97 (2.93, 3.01)	2.97 (2.93, 3.01)
Time-based HVS	6.18 (6.16, 6.20)	6.03 (6.00, 6.06)	33.34 (33.02, 33.66)	32.48 (32.18, 32.78)	3.15 (3.12, 3.18)	3.21 (3.19, 3.23)
Census-based HVS	6.15 (6.03, 6.27)	5.98 (5.91, 6.05)	33.04 (32.83, 33.25)	32.27 (31.98, 32.56)	3.12 (3.06, 3.18)	3.18 (3.13, 3.23)
Mixed HVS	6.17 (6.15, 6.19)	6.01 (5.98, 6.04)	33.20 (32.88, 33.52)	32.39 (32.10, 32.68)	3.14 (3.13, 3.15)	3.20 (3.17, 3.23)

Notes. * We consider four categories. ** For scenarios without HVS, we consider hospitalist visits based on the current system, which means that hospitalist visits are still part of the process.

As discussed above, scheduled hospitalist visits do not always translate into actual visits because their implementation depends on congestion in the corresponding ward. In the baseline specification, we assume that scheduled visits are implemented with probability 1 when ward occupancy is below the 75th percentile, with probability 0.75 when occupancy is between the 75th and 90th percentiles, and with probability 0.5 when occupancy is above the 90th percentile. To examine the sensitivity of our findings to these assumptions, we vary the implementation probabilities under moderate and high levels of inpatient congestion. Figure 6 summarizes the results. When all scheduled visits are implemented regardless of inpatient congestion, time-based scheduling reduces the impact of boarding congestion on treatment–section congestion by 35.16%, whereas census-based scheduling reduces it by 47.93%. Thus, in the absence of inpatient-side constraints, census-based scheduling outperforms time-based scheduling. As the implementation probabilities decrease, however, the mitigating effect of both policies becomes smaller. Moreover, the advantage of census-based scheduling gradually diminishes and eventually reverses, such that under severe inpatient congestion, time-based scheduling performs slightly better. These findings suggest that census-based scheduling is more sensitive to inpatient congestion because the visits it generates are more likely to coincide with periods of high ward occupancy. As a result, although census-based scheduling is more responsive to real-time boarding conditions and can deliver larger benefits when inpatient capacity is available, it may also create greater staffing and coordination pressure on the inpatient side during already congested periods. By contrast, time-based scheduling is less responsive to fluctuations in boarding demand, but it may be operationally more robust when inpatient services face severe resource constraints. Overall, the results highlight that the effectiveness of hospitalist visit scheduling depends critically on the ability of inpatient services to accommodate the scheduled visits and that the relative attractiveness of census-based versus time-based policies depends on the severity of inpatient congestion.

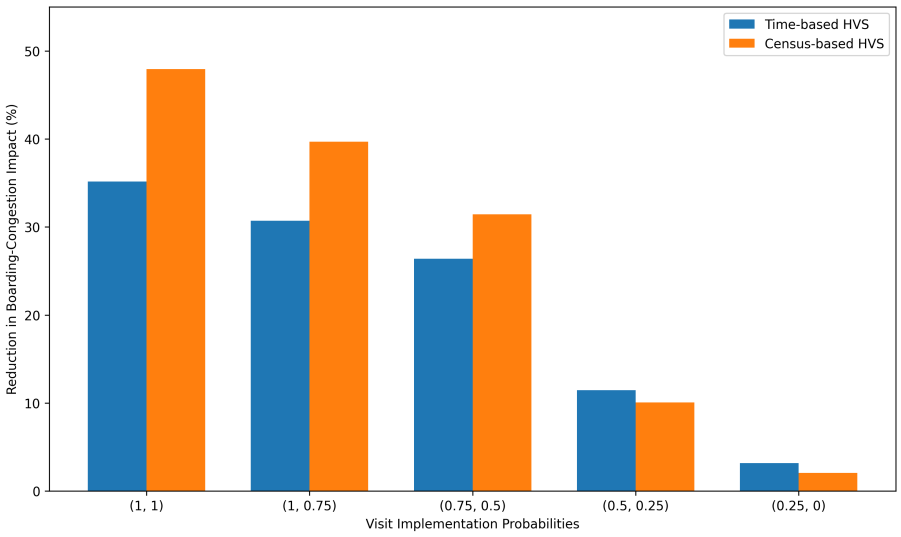


Figure 6 **Sensitivity of HVS Performance to Inpatient-Congestion Implementation Probabilities**

Finally, we evaluate the effectiveness of the proposed interventions across different ED operating conditions. We focus on three key characteristics of the ED: the arrival rate of emergency patients, treatment duration, and the likelihood of patient boarding (i.e., admission probability). For each characteristic, we consider three scenarios: a low scenario (25% below the base case), a medium scenario (the base case), and a high scenario (25% above the base case). Throughout this analysis, we maintain the baseline implementation probabilities for HVS under inpatient congestion, namely, 0.75 under moderate inpatient congestion and 0.5 under high inpatient congestion. Table 8 reports the resulting percentage reductions in the impact of boarding congestion on treatment–section congestion.

Observation 1. The effectiveness of all interventions generally increases as the arrival rate of emergency patients or treatment duration rises. This pattern indicates that the proposed interventions are especially valuable in EDs facing heavier operational pressure. For example, under low boarding probability, the reduction achieved by TP increases from 19.66% in the low-arrival/low-treatment scenario to 42.80% in the high-arrival/high-treatment scenario. Similar upward patterns are observed for both HVS policies. These results suggest that when demand is higher or treatment takes longer, boarding patients are more likely to create spillovers into the treatment section, and interventions that reduce this spillover generate larger benefits.

Observation 2. Across all scenarios, census-based HVS + TP performs best, followed by time-based HVS + TP, with TP alone delivering the smallest reduction. This finding suggests that hospitalist visits that respond directly to boarding demand remain the most effective policy, even when not all scheduled visits can be implemented because of inpatient congestion. For instance, in the baseline case with medium arrival, medium treatment, and medium boarding, TP reduces

the impact of boarding congestion by 26.86%, time-based HVS + TP increases this reduction to 32.83%, and census-based HVS + TP further increases it to 40.30%.

Observation 3. The effectiveness of all interventions declines as boarding probability increases, and the advantage of census-based HVS over time-based HVS becomes smaller under high boarding conditions. For example, in the high-arrival/high-treatment scenario, the gap between census-based and time-based HVS + TP decreases from 1.42 percentage points under low boarding probability (46.81% versus 45.39%) to 0.52 percentage points under high boarding probability (29.51% versus 28.99%). As boarding probability rises, downstream wards become more congested, which reduces the fraction of scheduled hospitalist visits that can actually be implemented. As a result, HVS remains beneficial, but its incremental value becomes smaller when inpatient congestion is more severe.

Table 8 Percentage Reductions in the Impact of Boarding Congestion on Treatment–Section Congestion Based on ED Characteristics

		TP			Time-based HVS + TP			Census-based HVS + TP		
Arrival	Treatment	Boarding percentage			Boarding percentage			Boarding percentage		
Rate	Time	Low	Med	High	Low	Med	High	Low	Med	High
Low	Low	19.66%	16.45%	7.81%	26.09%	24.82%	17.37%	32.21%	31.12%	19.56%
	Med	22.71%	18.05%	10.05%	33.18%	30.87%	20.52%	40.40%	37.65%	22.56%
	High	30.32%	21.46%	13.90%	37.17%	32.37%	24.99%	45.94%	39.70%	26.73%
Med	Low	24.45%	18.77%	14.03%	27.73%	25.98%	21.46%	34.25%	31.84%	23.64%
	Med	36.34%	26.86%	18.91%	37.65%	32.83%	23.55%	42.13%	40.30%	27.01%
	High	39.26%	28.61%	21.39%	41.51%	35.80%	27.62%	44.16%	43.56%	29.38%
High	Low	29.00%	23.65%	16.97%	32.83%	29.53%	22.05%	37.62%	35.76%	24.49%
	Med	39.85%	30.09%	21.21%	43.17%	33.65%	25.25%	45.27%	40.81%	27.46%
	High	42.80%	31.63%	22.31%	45.39%	36.27%	28.99%	46.81%	43.87%	29.51%

In summary, the simulation results show that both patient flow control and hospitalist visit scheduling can mitigate the adverse impact of boarding congestion on treatment–section congestion. Their benefits are greatest in EDs with higher arrival rates and longer treatment times, where boarding patients create greater spillovers into the treatment section. Census-based HVS consistently performs best, but its advantage over time-based HVS becomes smaller as boarding probability increases, because downstream inpatient congestion limits the feasibility of scheduled hospitalist visits.

6. Discussion and Conclusion

Using a comprehensive dataset spanning eight emergency departments, we examine the impact of boarding congestion on ED patient treatment time. Prior studies have largely treated boarding patients as occupying ED beds and potentially obstructing access for patients awaiting care, focusing

primarily on waiting time or length of stay as outcome measures. In contrast, we focus explicitly on treatment time and uncover a more nuanced relationship between boarding congestion and ED patient treatment time, characterized by an inverted U-shaped pattern. We find that an increase in boarding congestion from 10% to 40% increases the average treatment time by 19%. This positive relationship, however, holds only up to a threshold level of congestion, beyond which further increases in boarding congestion lead to shorter treatment times. We establish the robustness of these results through an extensive set of robustness checks.

We also investigate the mechanism underlying our findings. Because boarding congestion imposes additional workload on ED resources at low levels of congestion, boarding patients are primarily managed by ED staff. As a result, increases in boarding congestion lead to greater ED resource utilization and longer treatment times, explaining the upward-sloping portion of the inverted U-shaped relationship. However, at higher levels of boarding congestion, hospitalist involvement is more likely, shifting care responsibilities away from ED staff. This reallocation of care frees ED resources and alleviates capacity constraints in the treatment area, thus reducing ED treatment times and generating the downward-sloping portion of the inverted U-shaped relationship.

Based on empirical analyses, we propose two interventions to mitigate the impact of boarding congestion on treatment time: boarding patient flow control and hospitalist visit scheduling policies. The simulation results illustrate that implementing these two interventions together can reduce the effects of boarding congestion by approximately 44% for a tertiary hospital.

6.1. Implications for Theory

Our study has several implications for healthcare operations management theory. The first implication is for staff levels in EDs. Our study corroborates the descriptive findings in Armony et al. (2015) regarding the additional workload on the ED staff resulting from congestion caused by boarding patients. Although patient boarding is a common issue in EDs globally (Khare et al. 2009b), we do not find any OM/OR study quantifying the impact of the congestion caused by boarding patients on treatment time and considering it in ED staffing decisions. Previous studies have generally assumed that admitted patients promptly leave the ED, regardless of whether they are immediately transferred to the ward or remain in the ED awaiting admission. However, our findings demonstrate the significance of the boarding congestion effect, emphasizing the need for researchers to consider boarding patients' presence when planning staffing levels.

The second implication concerns patient flow in EDs. Previous research on patient flow generally divides patients into two categories: "new patients" awaiting treatment and "in-process patients" who are already receiving care. However, in our sample, 48.8% of boarding patients need extra ED resources beyond standard nursing checkups during their wait. This underscores the necessity of incorporating boarding patients into patient flow models in future ED studies.

Thirdly, our study holds implications for scholars interested in examining the role of hospitalists in EDs. While hospitalists play a crucial role in mitigating the adverse effects of boarding congestion on ED performance and patient outcomes (Chadaga et al. 2012), we are not aware of any OR/MS study that specifically focuses on this role or addresses operational questions related to it. This study proposes an easy-to-implement framework for selecting the optimal HVS policy for a given hospitalist based on their inpatient services' demand volume. This opens up a promising avenue of research on the role of hospitalists in the ED and its impact on both ED and hospital performance.

6.2. Implications for Practice

Our study also carries significant implications for practice. First, by analyzing patient–level data from eight EDs, our study highlights the importance of patient boarding in EDs and shows that the adverse impact of boarding congestion on the ED's performance is far beyond what was documented previously. These findings highlight the critical need for hospital administrators to carefully characterize boarding congestion in their ED operations. Ignoring this impact on treatment times when making operational decisions may result in persistent overcrowding and an inability to provide quality and timely care to patients. We note, however, that although our analysis controls for physician workload and shift-specific effects, physician-level heterogeneity within shifts may remain. Future research using physician-specific data could therefore examine how physician-level heterogeneity within shifts influences the relationship between boarding congestion and ED treatment times.

Financial incentives and the volatility of demand for ED services inevitably lead to the phenomenon of patient boarding in some EDs. This study proposes two easily implementable interventions to mitigate the negative impact of congestion caused by boarding patients on treatment time. These interventions are based on optimizing the use of ED and hospital resources and do not require major resource allocation or structural changes. ED managers can consider these interventions and customize them according to the size and complexity of the patient mix in their EDs to alleviate the adverse effects of boarding patient congestion on ED performance.

Our findings suggest that prioritizing boarding patients over emergency patients for ED resources can be effective. However, we show that this policy is not universally optimal for all EDs. Our sensitivity analyses in Section 5.2 indicate that prioritizing boarding patients is most effective for EDs characterized by (i) a high arrival rate of checkup requests from boarding patients, (ii) a high probability of emergency patients being admitted to the hospital, and (iii) a longer duration of checkup time for boarding patients in the treatment section. Overall, our results suggest that EDs dealing with a more acute and/or complex patient mix may benefit from prioritizing boarding patients, whereas EDs managing less acute or severe cases, such as community hospitals, may consider prioritizing emergency patients to improve their ED performance.

Regarding our second intervention, our analyses show that census-based HVS generally outperforms time-based HVS across a broad set of ED conditions, particularly when combined with TP. Its advantage is greater when arrival rates are higher or treatment durations are longer, as boarding patients create larger spillovers into treatment–section congestion under these conditions. However, the effectiveness of both HVS policies declines as boarding probability increases, because higher boarding levels are associated with greater inpatient congestion and a lower probability that scheduled hospitalist visits can be implemented. Consequently, the advantage of census-based HVS becomes smaller under high boarding conditions and may even disappear under severe inpatient-side constraints. These results suggest that census-based HVS is most effective in congested EDs when inpatient services still have enough operational flexibility to support scheduled visits. Future research can examine additional interventions to reduce boarding and investigate how inpatient-side resource constraints affect their effectiveness.

Our simulation does not model transportation and other transfer-related bottlenecks separately. Instead, these frictions are incorporated in reduced form through the observed boarding times in the data. Thus, the boarding delays represented in our model may reflect not only inpatient capacity constraints and hospitalist availability, but also downstream logistical factors such as transport staff availability, elevator congestion, bed cleaning delays, and handoff coordination. Future research with more granular operational data could explicitly model these transfer-related bottlenecks to examine how they interact with hospitalist scheduling and ED boarding.

Finally, the hospital admission team faces a dilemma when deciding whether to assign a patient to a nonprimary bed (off-service placement) when all primary beds are occupied or wait until a primary bed becomes available, thus leaving a nonprimary bed unused (Dai and Shi 2019). Given that off-service placement has negative impacts on patient outcomes and hospital length of stay, some studies recommend assigning patients to primary beds, even if it results in prolonged boarding time (Song et al. 2020). By quantifying the effects of boarding congestion on ED performance, our study provides valuable assistance to the hospital admission team in navigating this trade-off. However, further research is needed to develop a framework that guides practitioners in determining the optimal timing for off-service placement based on the current state of boarding and the hospital.

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Appendix: On the Effects of Boarding Congestion on Treatment Time in Emergency Departments

A1. Sample Construction and Descriptive Statistics

The initial dataset comprises 565,716 patient-level ED visits. As illustrated in Figure A.1, we apply several exclusion criteria to construct the analysis sample. First, we remove visits in which patients leave the ED before being seen by a physician, which account for approximately 5% of all visits; these cases are excluded because the treatment process is incomplete. Second, we exclude visits involving patients with CTAS levels 1 and 5, representing approximately 1.5% and 9.9% of total visits, respectively. The care processes for these patients differ substantially from standard ED workflows. CTAS 1 patients receive immediate, highest-priority care due to severe clinical risk and are typically managed in close coordination with inpatient services; once stabilized, they are transferred directly to inpatient care without entering the boarding stage. In contrast, CTAS 5 patients generally require minimal intervention and, in more than 98% of cases, interact with an ED physician only once after an extended wait period. Thus, the time they spend in the treatment section is likely less sensitive to operational conditions that affect ED resource availability. For completeness, we also conduct separate analyses including CTAS 1 and CTAS 5 patients; the results are reported in Appendix A2.7.

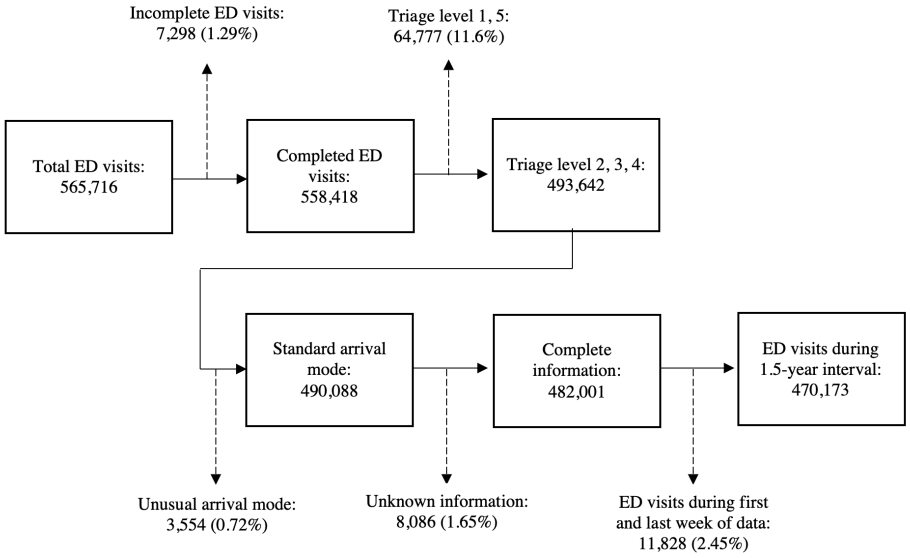


Figure A.1 Selection of the emergency department (ED) visit sample.

Next, we exclude patients who arrive by helicopter or are escorted by police, as different protocols are followed for these cases (less than 1% of observations). Additionally, we exclude ED visits with missing information (1.6% of observations). Finally, we exclude data from the first and last weeks

to avoid censored estimation of occupancy. This process results in a final sample of 470,173 patient-level ED visits across eight hospitals. Table A.1 demonstrates summary statistics for continuous and dummy variables for each hospital separately and Table A.2 provides the correlation matrix for the entire sample.

Table A.1 Summary Statistics

	Secondary			Tertiary				
Variable	H1	H2	H3	H4	H5	H6	H7	H8
TreatmentTime	323.69 (418.16)	385.17 (498.12)	310.69 (480.51)	372.80 (394.16)	406.76 (524.27)	320.31 (376.88)	434.12 (478.54)	471.33 (459.91)
Congestion	0.30 (0.16)	0.21 (0.16)	0.29 (0.20)	0.46 (0.13)	0.47 (0.17)	0.45 (0.15)	0.31 (0.12)	0.31 (0.15)
Age	51.93 (21.64)	48.67 (22.54)	49.18 (22.78)	53.41 (22.82)	49.65 (25.82)	52.72 (21.38)	50.90 (20.41)	49.81 (21.82)
Gender	0.37 (0.48)	0.47 (0.50)	0.46 (0.50)	0.45 (0.50)	0.45 (0.50)	0.44 (0.50)	0.45 (0.50)	0.50 (0.50)
ArrMode	0.77 (0.42)	0.70 (0.46)	0.80 (0.40)	0.68 (0.47)	0.71 (0.45)	0.82 (0.39)	0.75 (0.44)	0.61 (0.49)
Triage	3.39 (0.63)	3.24 (0.70)	3.61 (0.57)	2.98 (0.73)	2.99 (0.75)	3.09 (0.74)	3.19 (0.72)	3.11 (0.77)
WaitCensus	0.18 (0.11)	0.25 (0.14)	0.24 (0.16)	0.20 (0.09)	0.24 (0.10)	0.22 (0.11)	0.21 (0.12)	0.12 (0.07)
TreatmentCensus	0.48 (0.13)	0.48 (0.11)	0.41 (0.14)	0.50 (0.11)	0.56 (0.12)	0.55 (0.13)	0.56 (0.13)	0.50 (0.12)
Workload	4.46 (1.21)	4.10 (0.96)	1.90 (0.64)	6.94 (1.55)	8.11 (1.73)	10.07 (2.43)	6.06 (1.35)	5.91 (1.37)
AvgTriage	3.56 (0.13)	3.56 (0.14)	3.78 (0.11)	3.24 (0.11)	3.29 (0.10)	3.30 (0.12)	3.39 (0.13)	3.21 (0.15)
HospOccupancy	0.85 (0.05)	0.84 (0.07)	0.81 (0.08)	0.84 (0.05)	0.88 (0.05)	0.89 (0.04)	0.84 (0.07)	0.83 (0.05)
Weekend	0.25 (0.43)	0.28 (0.45)	0.27 (0.44)	0.26 (0.44)	0.24 (0.42)	0.25 (0.44)	0.26 (0.44)	0.26 (0.44)
Sample size (N)	50362	39302	23688	72684	78154	111693	50742	43548

Notes. †: Waiting room census and treatment section census are normalized based on the capacity of each ED. The first number is mean of the variable, and standard errors are shown in parentheses.

Table A.2 Correlation Matrix of continuous variables

	TreatmentTime	*Congestion*	*Age*	*Gender*	*ArrMode*	*Triage*	*WaitCensus*	*TreatmentCensus*
TreatmentTime	1.00							
Congestion	-0.03	1.00						
Age	0.31	0.03	1.00					
Gender	0.01	-0.01	-0.00	1.000				
ArrMode	-0.34	0.03	-0.30	-0.02	1.00			
Triage	-0.18	-0.09	-0.02	-0.03	0.13	1.00		
WaitCensus	0.01	0.13	0.04	-0.02	0.06	-0.05	1.00	
TreatmentCensus	-0.01	0.18	0.04	-0.01	0.06	-0.04	0.23	1.00
Workload	-0.02	0.39	0.05	-0.01	0.07	-0.12	0.18	0.74
AvgTriage	-0.04	-0.30	-0.02	-0.02	0.04	0.23	0.06	-0.17
HospOccupancy	-0.01	0.36	0.00*	-0.01	0.03	-0.06	0.03	0.12
Weekend	-0.01	-0.16	-0.03	0.01	-0.02	-0.00*	-0.15	-0.32

	Workload	*AvgTriage*	*HospOccupancy*	*Weekend*
Workload	1.00			
AvgTriage	-0.46	1.00		
HospOccupancy	0.32	-0.17	1.00	
Weekend	-0.20	-0.05	-0.05	1.00

Note. Except the ones with the symbol "*", all correlations are significant at 5% significance level.

A2. Robustness Checks

We perform a series of robustness checks to evaluate the sensitivity of our main results to alternative analytical assumptions, model specifications, and competing theoretical mechanisms.

A2.1. Alternative Operationalization of Boarding Congestion

We conduct three robustness analyses using alternative measures of boarding congestion determined at the onset of treatment. Our period-average measure raises two related concerns. First, because the treatment period determines the averaging window, *Congestion* may be mechanically correlated with *TreatmentTime*, raising the possibility that the observed relationship is partly driven by this mechanical relationship rather than by the underlying effect of congestion. Second, the period-average measure incorporates congestion information realized after treatment has begun and therefore is not available when treatment initially commenced, potentially weakening the causal interpretation of the relationship between congestion and treatment time. By measuring congestion at treatment onset, our alternative measures address both concerns: they are determined before subsequent treatment unfolds and therefore neither mechanically depend on the realized treatment duration nor incorporate future congestion realizations.

First, we measure Congestion using the same boarding capacity definition as in our main analysis—the 95th percentile of hourly boarding census for each ED—but measure boarding census at the onset of treatment rather than averaging it over the treatment period. This measure retains the normalization used in our main analysis while eliminating the potential mechanical relationship between *Congestion* and *TreatmentTime* and the incorporation of future congestion information. Second, we measure *Congestion* as the ratio of boarding census to the number of licensed ED beds at the onset of treatment for the patient in ED visit i at hospital h. This measure is motivated by Batt and Terwiesch (2017), who use the waiting room census at treatment initiation to capture the incremental workload imposed on ED staff by queued patients. Because boarding census varies considerably across EDs, we normalize this measure within each hospital. Third, we construct a time-based measure of boarding congestion at treatment onset, defined as the average remaining boarding time of all boarding patients at that point. This measure is motivated by Carmen et al. (2018), who analytically show that ED performance can be affected not only by the number of boarding patients but also by the duration for which they remain boarded. We again normalize this measure within each ED.

We re-estimate Equations 2 and 3 using each alternative measure. The results, reported in Table A.3, continue to exhibit a robust quadratic relationship between *Congestion* and *TreatmentTime*. These results indicate that our findings are robust to alternative measures of congestion based on information available at treatment onset and are therefore unlikely to be driven by either the mechanical relationship inherent in the period-average measure or the incorporation of future congestion realizations. Importantly, while these onset measures provide a more conservative test of the relationship, their consistency with our period-average measure establishes that our

Table A.3 Estimation Results with Alternative Variables for Boarding Congestion

	(1) Based on 95th percentile		(2) Based on licensed beds		(3) Based on total remaining time	
Congestion	0.68***	(0.055)	0.59***	(0.132)	0.43**	(0.139)
Congestion^2	-1.04***	(0.043)	-0.93***	(0.158)	-0.72***	(0.199)
Controls	Included		Included		Included	
Sample size	470,173		470,173		470,173	
Ancillary parameter σ	1.07		1.07		1.07	
Ancillary parameter κ	0.98***		0.98***		0.98***	
Wald χ^2	134,749***		135,113***		134,629***	

Notes. Clustered robust standard errors in parentheses. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

key independent variable remains appropriate for capturing the theoretical congestion environment experienced throughout the treatment period without compromising identification.

In addition to these alternative onset measures, we conduct a complementary robustness analysis that explicitly models the treatment process as a duration process with time-varying boarding congestion. Specifically, we estimate a discrete-time hazard model in which each treatment episode is divided into time intervals, with treatment completion in a given interval modeled as a function of the boarding congestion observed up to that point. This approach preserves the temporal variation in congestion within treatment episodes and avoids defining congestion using information realized after treatment completion. The results, reported in Table A.4, yield a nonlinear relationship between time-varying congestion and the probability of treatment completion that is consistent with our main findings. Specifically, the completion hazard exhibits the corresponding U-shaped relationship with congestion, implying a corresponding inverted-U relationship between congestion and treatment duration. Thus, the nonlinear relationship persists when congestion is allowed to vary dynamically over the treatment episode, providing evidence that our main result is not an artifact of averaging congestion over the treatment episode. Taken together with the results from the alternative onset measures, this analysis further supports our use of period-average congestion as a parsimonious measure of the overall congestion environment experienced during treatment.

Table A.4 Estimation Results with Discrete-Time Hazard Model

	(1) Baseline Model		(2) Control Function	
Congestion	-0.27***	(0.005)	-0.26***	(0.005)
Congestion^2	0.27**	(0.005)	0.27***	(0.005)
Controls	Included		Included	
Sample size	2,325,582		2,325,582	
Wald χ^2	5,107***		5,113***	

Notes. Clustered robust standard errors in parentheses. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

A2.2. Patient Heterogeneity

In our main analysis, we estimate the average effect of boarding congestion on treatment time across the full patient sample. However, treatment time may vary systematically by diagnostic category, raising the possibility that the observed inverted U-shaped relationship reflects patient heterogeneity rather than the effect of boarding congestion itself. To assess this alternative explanation, we conduct a split-sample analysis in which we re-estimate our main specifications (Equations 2 and 3) for each diagnostic code separately. The results, reported in Table A.5, are consistent with our baseline findings: the linear term of *Congestion* is positive and the quadratic term is negative across all diagnostic codes. Moreover, the quadratic coefficients are statistically significant in every subsample. These findings indicate that the nonlinear relationship between boarding congestion and treatment time is robust across diagnostic categories, ruling out patient heterogeneity as an alternative explanation.

Table A.5 Boarding Congestion's Effect on Treatment Time across Diagnostic Codes

Diagnostic code	Sample size	*Congestion*		$Congestion^2$	
1	32,443	0.96^{**}	(0.322)	-1.12^{***}	(0.289)
2	24,192	0.68	(0.353)	-1.41^{***}	(0.337)
3	32,264	0.56	(0.290)	-0.97^{***}	(0.288)
4	25,201	0.68^{*}	(0.278)	-1.22^{***}	(0.316)
5	36,527	0.51^{**}	(0.126)	-1.02^{***}	(0.273)
6	12,930	0.73^{**}	(0.252)	-1.18^{**}	(0.429)
7	7,119	0.65	(0.501)	-1.03^{***}	(0.308)
8	16,474	0.71	(0.473)	-1.11^{*}	(0.443)
9	12,985	1.37^{*}	(0.614)	-1.27^{*}	(0.547)
11	32,422	0.47	(0.249)	-0.69^{**}	(0.221)
12	73,235	0.47^{***}	(0.135)	-0.82^{***}	(0.185)
13	138,964	0.73^{***}	(0.128)	-1.02^{***}	(0.214)
†14	25,417	0.37	(0.445)	-0.90^{*}	(0.420)

Notes. For brevity, only the coefficients for *Congestion* and $Congestion^2$ are displayed. †: Samples from five diagnostics codes are combined due to the small sample size. Clustered robust standard errors in parentheses. $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

A2.3. Hospital Heterogeneity

A further source of potential heterogeneity arises at the hospital level. If boarding congestion systematically differs across hospitals, the estimated quadratic relationship could reflect cross-hospital differences, rather than the effect of congestion itself. To evaluate this possibility, we conduct an additional split-sample analysis in which we re-estimate our main specifications (Equations 2 and 3) separately for each hospital. As reported in Table A.6, the results remain qualitatively unchanged: the quadratic term is statistically significant in every hospital-specific subsample. These findings indicate that the nonlinear relationship between boarding congestion and treatment time is robust across hospitals, effectively ruling out hospital-level heterogeneity as an alternative explanation.

Table A.6 Boarding Congestion's Effect on Treatment Time across Hospitals

Hospital	Sample size	*Congestion*		*Congestion*^{2}	
1	50,362	0.53^{*}	(0.210)	-1.56^{***}	(0.160)
2	72,684	1.64^{***}	(0.226)	-2.75^{***}	(0.150)
3	78,154	2.09^{***}	(0.232)	-2.23^{***}	(0.154)
4	111,693	0.92^{***}	(0.278)	-1.00^{***}	(0.120)
5	39,302	0.29	(0.291)	-0.88^{***}	(0.220)
6	50,742	1.09^{***}	(0.253)	-1.68^{***}	(0.202)
7	43,548	0.82^{***}	(0.218)	-1.36^{***}	(0.154)
8	23,688	0.68^{***}	(0.195)	-1.16^{***}	(0.182)

Notes. For brevity, only the coefficients for *Congestion* and *Congestion*^{2} are displayed. Robust standard errors in parentheses. $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

A2.4. Reverse Causality

One may also be concerned about reverse causality between $Congestion$ and $TreatmentTime$, whereby boarding congestion affects ED treatment time, while ED treatment time simultaneously contributes to boarding congestion. Our control function approach effectively addresses this concern (Wooldridge 2015). Specifically, our instrumental variable–the utilization rate of the nearest inpatient service–is orthogonal to treatment time in the focal ED; that is, a patient's treatment time at the focal hospital's ED does not depend on utilization levels at a nearby hospital's inpatient service. This exclusion restriction allows us to recover an unbiased estimate of the causal effect of boarding congestion on treatment time.

Regardless, we further examine the causal effect of boarding congestion on treatment time using an alternative specification that explicitly incorporates simultaneity. Specifically, we estimate a simultaneous equations system using three-stage least squares (3SLS), which accounts for the endogeneity of both $Congestion$ and $TreatmentTime$ and exploits cross-equation correlation in the error terms. The 3SLS estimator combines instrumental variables with seemingly unrelated regressions, yielding consistent and efficient estimates of the structural parameters. Identification is achieved through the exclusion of the instrumental variable, $NearestIP$, from the $TreatmentTime$ equation, with $NearestIP$ included only in the $Congestion$ equation. Patient-, ED-, and hospital-level control variables, along with seasonality and time-of-day fixed effects, are included in both equations. Because 3SLS is fundamentally a linear method, we define $TreatmentTime$ for ED visit i at hospital h as the natural logarithm of the elapsed time (in minutes) from when the patient is called to the treatment section until the physician makes the disposition decision. Table A.7 reports the results from this estimation. As expected, we find that longer treatment times (i.e., slower patient flow from the ED to the boarding area) lead to lower boarding congestion ($\beta=-1.81$, $p<0.001$). Importantly, even after accounting for simultaneity, the estimated effect of boarding congestion on treatment time remains qualitatively unchanged ($\beta_1=1.29$, $p=0.005$ and $\beta_2=-2.04$, $p<0.001$).

Table A.7 3SLS Estimation Results

	$TreatmentTime$	$Congestion$
$Congestion$	1.29** (0.466)	
$Congestion^2$	-2.04*** (0.519)	
$TreatmentTime$		-1.81*** (0.047)
$NearestIP$		0.08*** (0.019)
Controls	Included	Included
Wald χ^2	148,284***	2,497***
Sample size	470,173	

Notes. $^*p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

A2.5. Potential Regional Demand Spillovers

A potential concern with our instrumental-variable strategy is that utilization at the nearby inpatient service may reflect regional demand shocks that simultaneously affect treatment time at the focal ED through channels other than boarding congestion, which can undermine the exogeneity assumption for the instrument variable. To examine the plausibility of this concern, we re-estimate our control function specification while including calendar-date fixed effects. These fixed effects absorb arbitrary shocks common to patients treated across hospitals on the same day, including regional fluctuations in healthcare demand, thereby allowing identification to rely on variation in inpatient capacity conditions beyond common date-level shocks. As reported in Table A.8, the results are highly consistent with our main estimates, suggesting that common regional demand shocks are unlikely to undermine the exogeneity assumption for our instrument variable.

Table A.8 Estimation Results with Calendar-Date Fixed Effects

	First Stage	Control Function
	$Congestion$ (1)	$ln(TreatmentTime)$ (2)
$NearestIP$	0.18*** (0.018)	
$Congestion$		0.82*** (0.192)
$Congestion^2$		-1.09*** (0.214)
$Calendar-date\ FE$	Included	Included
$Other\ controls$	Included	Included
Sample Size	470,173	470,173

Notes. Clustered robust standard errors in parentheses. $^*p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

A2.6. Diagnostic Analysis with Respect to Unobservables

The exogeneity assumption of our instrument can only be justified theoretically, as it cannot be tested directly. We argue that our instrumental variable—the utilization rate of the nearest inpatient service—likely satisfies this assumption, since inpatient services at nearby hospitals do not

share resources with the focal hospital's ED (as confirmed by partner hospitals) and, therefore, are unlikely to affect treatment time in the focal ED. However, if there exists any unobserved factor that simultaneously affects both boarding congestion in the focal hospital and the utilization of the nearest inpatient service, the exogeneity assumption would be violated, potentially biasing the estimated effect of boarding congestion despite using the control function approach.

To assess this concern, we implement the diagnostic method proposed by Altonji et al. (2005), which compares the estimated coefficient from the full model (i.e., β_F–as estimated in Table 1) with that from a restricted model excluding control variables (i.e., β_R) through the ratio $\rho = \beta_F/(\beta_F - \beta_R)$. This ratio measures the relative influence of unobservables compared to observables in accounting for the results from the full model. According to Altonji et al. (2005), $|\rho| > 1$ indicates that the effect of unobservables is not strong enough to overturn the qualitative insights derived from the full model. In our analysis, $|\rho| = 7.59$ for $Congestion$ and $|\rho| = 2.40$ for $Congestion^2$, implying that unobservables would have to be at least 2.40 times stronger than observables to alter the main findings. Since both values exceed the threshold of 1, we conclude that our control function estimates are robust to potential omitted variables.

A2.7. Excluded CTAS Categories

Our main analysis excludes CTAS 1 and CTAS 5 patients due to their distinct ED workflows. For completeness, we include these patients and conduct a split-sample analysis for each CTAS level, which also allows us to examine the role of triage-related patient heterogeneity. As shown in Table A.9, the results for CTAS 2, 3, and 4 patients remain qualitatively unchanged, confirming that the nonlinear relationship between $Congestion$ and $TreatmentTime$ is robust across these triage levels.

For CTAS 1 and 5 patients, neither the linear ($\beta_1 = 0.05$, $p = 0.95$ for CTAS 1 patients and $\beta_1 = 0.19$, $p = 0.49$ for CTAS 5 patients) nor the quadratic term ($\beta_2 = 0.17$, $p = 0.79$ for CTAS 1 patients and $\beta_2 = -0.05$, $p = 0.56$ for CTAS 5 patients) of $Congestion$ is statistically significant. Even under a linear specification, the effect of congestion remains insignificant ($\beta = 0.23$, $p = 0.12$ for CTAS 1 patients and $\beta = 0.14$, $p = 0.53$ for CTAS 5 patients). This is expected, as high-priority CTAS 1 patients are stabilized and transferred directly to inpatient care, making their treatment time largely insensitive to ED operational conditions such as boarding congestion. Similarly, low-acuity patients' treatment time is less affected by operational factors such as ED boarding congestion.

Table A.9 Boarding Congestion's Effect on Treatment Time across Triage Levels

Triage level	Sample size	*Congestion*		*Congestion*^{2}	
1	8,523	0.05	(0.714)	0.17	(0.639)
2	100,262	0.89^{***}	(0.246)	-1.12^{***}	(0.298)
3	204,275	0.61^{**}	(0.232)	-1.04^{***}	(0.264)
4	165,636	0.66^{***}	(0.122)	-0.96^{***}	(0.133)
5	56,254	0.19	(0.267)	−0.05	(0.092)

Notes. For brevity, only the coefficients for *Congestion* and *Congestion*^{2} are displayed. Clustered robust standard errors in parentheses. $^*p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$.

A2.8. Alternative Theoretical Mechanisms

In our main analysis, we demonstrate that our empirical findings are driven by the level of ED resource utilization by boarding patients and how this utilization changes by whether boarding patients are primarily managed by ED staff versus hospitalists. Here, we explore three alternative mechanisms that might explain the relationship between boarding congestion and ED patient treatment times.

A2.8.1. Waiting Room Congestion: Prior research shows that ED boarding congestion affects length of stay and waiting times for ED patients (Khare et al. 2009a, White et al. 2013, Carmen et al. 2018). These studies suggest that boarding congestion might impede the smooth flow of newly arriving patients into the treatment areas. Moreover, Batt and Terwiesch (2017) document an inverted U-shaped relationship between waiting room congestion and ED patient treatment times. Accordingly, one might argue that the inverted U-shaped relationship we identify between boarding congestion and ED treatment times is driven by boarding congestion's effect on waiting room congestion. Under this explanation, severe delays in transferring new patients from the waiting room to treatment areas increase pressure on ED physicians to accelerate care for patients already being treated.

This mechanism implies that waiting census mediates the relationship between boarding congestion and ED patient treatment times. To examine this possibility, we re-estimate Equation 4 and augment it with the following equation:

$$WaitCensus_{ih} = \eta_0 + \eta_1 Congestion_{ih} + \mathbf{Controls'}_{ih}\boldsymbol{\eta} + \varrho_{ih} \tag{A.1}$$

The results show that the estimated coefficient on *Congestion* in the *WaitCensus* equation is not statistically significant ($\eta_1 = 0.004$, $p = 0.709$). Thus, we find no evidence that boarding congestion is associated with waiting room congestion, ruling out this alternative explanation for the observed inverted U-shaped relationship.

A2.8.2. Accelerating Care: Prior literature suggests that service rates can respond to system load (Kc and Terwiesch 2009). For example, when physicians face increased time pressure–such as when appointments approach the end of the day or delays accumulate–they may accelerate care to complete all scheduled visits (Freedman et al. 2021). By analogy, because boarding congestion represents an additional source of system load for ED physicians, one might expect physicians to speed up treatment for ED patients when boarding congestion is high.

To test this alternative mechanism, we construct a new frequency variable, $DischargedPatients_{ih}$, defined as the number of patients discharged from the ED per minute during the treatment time of the patient in ED visit i at hospital h. Higher values of $DischargedPatients_{ih}$ indicate accelerated care. This mechanism implies a U-shaped relationship between boarding congestion and $DischargedPatientss_{ih}$, and that $DischargedPatients_{ih}$ mediates the relationship between boarding congestion and ED patient treatment times. To examine this possibility, we re-estimate Equation 4 and augment it with the following specification:

$$DischargedPatients_{ih} = \eta_0 + \eta_1 Congestion_{ih} + \eta_2 Congestion_{ih}^2 + \mathbf{Controls'}_{ih}\boldsymbol{\eta} + \varrho_{ih} \tag{A.2}$$

The results show that neither the linear term for $Congestion$ ($\eta_1 = -0.14$, $p = 0.200$) nor the quadratic term ($\eta_2 = 0.37$, $p = 0.201$) is statistically significant in the $DischageredPatients$ equation. Thus, we find no evidence that boarding congestion is associated with faster treatment by ED physicians, ruling out this second alternative explanation for the observed inverted U-shaped relationship.

A2.8.3. Change in Ordering Diagnostic Tests for ED Patients: One may also argue that when ED physicians face increased time pressure due to boarding congestion, they may be more likely to order diagnostic tests for patients undergoing treatment in order to reduce face-to-face time. Under this view, the relationship we observe between boarding congestion and ED patient treatment times may be driven by changes in physicians' diagnostic test ordering behavior.

To test this alternative mechanism, we construct a new frequency variable, $DiagnosticTests_{ih}$, defined as the number of diagnostic tests ordered per minute during the treatment time of the patient in ED visit i at hospital h. We then re-estimate Equation 4 and augment it with the following specifications, estimated one at a time:

$$DiagnosticTests_{ih} = \eta_0 + \eta_1 Congestion_{ih} + \mathbf{Controls'}_{ih}\boldsymbol{\eta} + \varrho_{ih} \tag{A.3}$$

$$DiagnosticTests_{ih} = \eta_0 + \eta_1 Congestion_{ih} + \eta_2 Congestion_{ih}^2 + \mathbf{Controls'}_{ih}\boldsymbol{\eta} + \varrho_{ih} \tag{A.4}$$

The results indicate no evidence of either a linear relationship between $Congestion$ and $DiagnosticTests$ ($\eta_1 = 0.02$, $p = 0.259$ in Equation A.3) or a quadratic relationship ($\eta_1 = 0.0003$,

$p = 0.992$ and $\eta_2 = 0.019$, $p = 0.644$ in Equation A.4). Thus, we find no evidence that boarding congestion is associated with the frequency of diagnostic test ordering by ED physicians, ruling out this third alternative explanation for the observed inverted U-shaped relationship.

A2.8.4. Operationalization of Hospitalist Visit: Similar to the potential mechanical correlation between $Congestion$ and $TreatmentTime$, one could argue that $HospitalistVisit$ may also be mechanically correlated with $TreatmentTime$ because we define $HospitalistVisit$ based on whether a hospitalist visit occurs during the treatment interval. Such a correlation could raise the concern that the observed relationship between $Congestion$ and $HospitalistVisit$ reflects the effect of $TreatmentTime$, which mechanically affects the opportunity for a hospitalist visit, rather than a mechanism through which congestion induces hospitalist involvement.

To explore this possibility, we include $TreatmentTime$ directly in the $Pr(HospitalistVisit_{ih} = 1)$ equation in Equation 4. The estimated coefficients on $Congestion$ ($\alpha_1 = 2.64$, $p < 0.001$) and $Congestion^2$($\alpha_2 = -1.69, p < 0.001$) remain virtually unchanged. This suggests that the relationship between congestion and hospitalist involvement empirically remains the same even after accounting for the potential mechanical correlation between $HospitalistVisit$ and $TreatmentTime$, providing evidence against this alternative explanation.

A3. Care by Hospitalists versus ED Physicians

In this section, we provide empirical evidence for two theoretical arguments that inform our proposed underlying mechanism. First, we examine how hospitalists' assumption of care responsibility affects boarding patients' need for routine checkups from ED physicians. Second, we examine how visits by ED physicians versus hospitalists influence the propensity to order diagnostics tests for boarding patients. Because both arguments pertain specifically to boarding patients, we restrict our analysis to the 73,930 patients who were admitted to an inpatient service and therefore experienced the boarding stage.

To assess the effect of hospitalist visits on the likelihood that boarding patients require attention from ED physicians, we estimate the following probit model:

$$Pr(CheckUp_{ih} = 1) = Pr(\beta_0 + \beta_1 Hospitalist_{ih} + \mathbf{Controls}_{ih}\boldsymbol{\beta} + \epsilon_{ih} > 0) \tag{A.5}$$

where $CheckUp_{ih}$ is a binary variable indicating whether the patient in ED visit i at hospital h received a routine checkup from an ED physician during boarding, and $Hospitalist_{ih}$ is a binary variable indicating whether the patient was visited by a hospitalist during boarding. The estimation of Equation A.5 shows that boarding patients who are visited by a hospitalist are less likely to require a routine checkup from an ED physician ($\beta_1 = -0.03$, $p = 0.019$).

Next, we compare the impact of visits by ED physicians versus hospitalists on the propensity to order diagnostic tests for boarding patients using the following probit model:

$$Pr(Test_{ih}=1)=Pr(\beta_0+\beta_1 EDPhysician_{ih}+\beta_2 Hospitalist_{ih}+\mathbf{Controls}_{ih}\boldsymbol{\beta}+\epsilon_{ih}>0) \quad \text{(A.6)}$$

where $Test_{ih}$ is a binary variable indicating whether the patient in ED visit i at hospital h received a diagnostic test order during boarding, and $EDPhysician_{ih}$ is a binary variable indicating whether the patient was visited by an ED physician during boarding. We find that ED physician visits are associated with a higher likelihood of diagnostic test ordering ($\beta_1=0.04$, $p<0.001$), whereas hospitalist visits are associated with a lower likelihood of test ordering ($\beta_2=-0.04$, $p<0.001$). A Wald test comparing the two coefficients confirms that hospitalists are significantly less likely to order diagnostic tests for boarding patients than ED physicians ($\chi^2=30.35$, $p<0.001$).

A4. Simulation Model

A4.1. Simulation Parameters

Table A.10 Simulation Paramters

Type	Description	Symbol	Value
Service rates (persons per hour)	Processing of orders	μ_3	3.4
	Processing of boarding patients with ED physician	$\mu_{b,2}$	1.4
	Processing of boarding patients with hospitalist	μ_h	0.9
Workers and beds	Processing of orders	m_3	15
	Beds	n	46
	ED physician capacity	s_{2-4}	8

Note that this table includes parameters calibrated using simulation and fixed variables. Other variables, including the number of orders per patient, patient arrival rate, and patient-physician service time in phases 2 and 4, are sampled directly from the data.

Table A.11 Patient Flow Control Policy Parameters Based on Changes in ED Characteristics

	High				Low			
	Parameters (L_i^l, L_i^m, L_i^h)			Patients*	Parameters (L_i^l, L_i^m, L_i^h)			Patients*
	L_1	L_2	L_3		L_1	L_2	L_3	
Patient arrival rate	(33, 38, 42)	(25, 30, 33)	(3, 6, 7)	Treatment**	(38, 40, 45)	(28, 33, 35)	(3, 4, 5)	Boarding
Checkup request rate	(38, 40, 45)	(28, 31, 35)	(3, 4, 5)	Boarding	(34, 39, 45)	(26, 30, 33)	(3, 5, 6)	Treatment
Admission probability	(35, 40, 45)	(27, 30, 35)	(2, 4, 5)	Boarding	(35, 38, 43)	(25, 31, 34)	(3, 5, 6)	Treatment
Processing of boarding patients†	(35, 38, 43)	(26, 28, 33)	(3, 5, 6)	Treatment	(35, 40, 45)	(26, 32, 35)	(2, 4, 5)	Boarding
Processing of treatment patients	(35, 41, 45)	(28, 33, 35)	(3, 4, 5)	Boarding	(33, 38, 42)	(23, 28, 34)	(3, 5, 5)	Treatment

Note: "High" indicates a 25% increase in the ED characteristic of interest, and "Low" denotes a 25% decrease in the ED character of interest. *: Patients refers to the type of patients prioritized more compared to the base model in which L_1: (35, 39, 45), L_2: (26, 30, 34), and L_3: (3, 5, 5). **: Treatment patients refers to patients currently undergoing treatment in the treatment section.
†: Processing of boarding patients and emergency patients in the treatment section by an ED physician which is based on persons per hour. A higher number indicates shorter visit times.

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